SML201 Chapter 2.2 *

Exploratory Data Analysis: Part I

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R functions covered

- Arithmetic: `diff()`
- Check and remove objects in the environment: `objects()`, `ls()`, `rm()`
- Save and load variables: `save(..., file=...)`, `load(..., file=...)`
- Data extraction: `$` by name; `[,]` by indices; with conditions or logical arguments
- Combining conditions: `&`, `|`, `!`
- Data manipulation: `na.omit()`, `sort(..., decreasing = FALSE)`, `unique()`, `rank()`, `%in%`
- Statistic functions: `sd(..., na.rm = ...)`, `var(..., na.rm = ...)`, `quantile(..., na.rm =..., p=...)`, `IQR(..., na.rm =...)`
- {dplyr} functions: see R videos (e.g. `select()`, `filter()`, `slice()`, `slice_max()`, `slice_min()`, `distinct()`, `count()`, `group_by()`, `summarize()`, `arrange()`, `n()` etc.)

Introduction

In the last chapter we introduced the data frame, an object type in R. We also introduced various statistics that summarize one-dimensional data. We hope you are convinced that these statistics are useful for extracting interesting information from a dataset. We will now go into more detail to show you how to calculate other more advanced statistics and how to use R to obtain them. We will also show you how to transform data with the package dplyr.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':
```

```
    intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

Some helpful notes

We'll begin with some information that will help you navigate your workspace in R.

Clearing variables in your workspace (i.e., global environment)

```
ls()  # check what R objects are currently in your workspace
```

```
character(0)
```

```
rm(list = ls())  # remove all the R objects in your workspace

x = c(3, 1, 4)  # create a variable x that stores the vector c(3,1,4)
y = 1:2
rm(x)  # remove just x from the workspace

x = c(3, 1, 4)
y = 1:2
rm(list = c("x", "y"))  # remove x and y from the workspace

x = c(3, 1, 4)
y = 1:2
rm(list = ls())
# remove all R objects in your workspace same as
# rm(list=c('x', 'y')) since x and y are the only R objects
# in your workspace
```

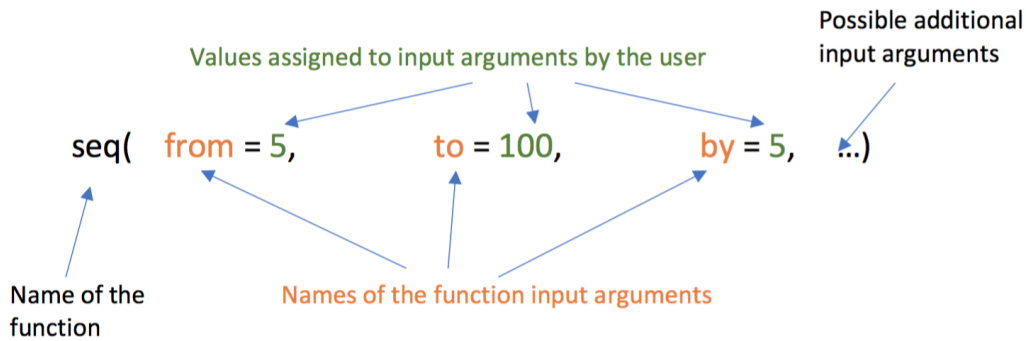
To run a line of code from the script file use

- `Command + Return` for Mac
- `Control + Return` for PC

You can see all the short-cut keys here https://support.rstudio.com/hc/en-us/articles/200711853-Keyboard-Shortcuts

Functions in R

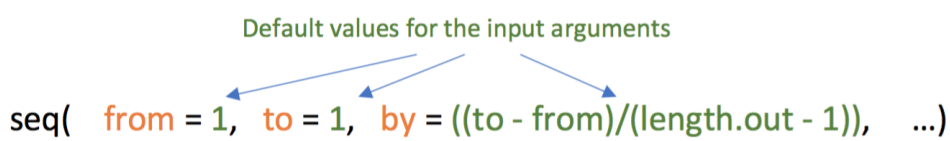
A function in R takes some data (i.e., a column or multiple columns in the dataset), manipulates the data (e.g., sum up of, take the maximum of, or make a graph with the data), and outputs the desired result. A function in R has the following format:



Input arguments are separated by commas.

Reading the help manual for a function

On the help manual of a function if there is already a default value assigned to the input argument, this means that if the user does not assign a value to the input argument the default value will be used by `R`.



For example,

```
seq()
```

```
[1] 1
```

Save and load specific variables

```
x = c(3, 1, 4)
y = 1:2

save("x", "y", file = "xy.RData")  # save x, y variables to xy.RData
# xy.RData will be saved to current work directory

save("x", "y", file = "xy.RData")
# can also specify a full path to tell R where to save
# variable x and y

rm(list = c("x", "y"))
```

```
# remove x and y from the workspace to see if save() works

ls()
```

```
character(0)
```

```
load(file = "xy.RData")  # load x, y back
ls()  # check if we get x and y back
```

```
[1] "x" "y"
```

```
x
```

```
[1] 3 1 4
```

```
y
```

```
[1] 1 2
```

```
rm(list = ls())
```

Content Overview

Today we'll be looking at data from the Translation Database, an open-source project that attempts to track all fiction, poetry, children's books, and nonfiction translated into English and published in the United States since 2008. The project was created by the literary hub Three Percent, in collaboration with Open Letter Books at the University of Rochester. The name "Three Percent" is derived from the oft-cited statistic that about 3% of all books published in the US are translations. In many parts of the globe, this figure is much higher, hovering around 30, 40, or even 50% or more.

Recent scholarship has been approaching controversial topics such as translation across race and gender. Scholars, writers and translators alike must be familiar with the metadata around publishing: which writers are being translated? From what countries? From what languages? Who is doing the translating? Who is doing the publishing? VIDA, an organization supporting women in the literary arts, has taken up a similar analysis with original writing, showing the numeric breakdown between men and women published in major literary journals and magazines. The inequities are startling.

Even if translation is not a primary component of your research, it no doubt affects your work. Translation is one of the primary mechanisms through which knowledge is exchanged worldwide. It is likely that many of the research articles or even primary sources that you are reading have been translated from other languages into English.

For today's lesson, let's first read in the dataset.

```
library(readr)
translations = read_csv(file = "~/Documents/Princeton/CDH/SML 201/Module/TranslationDatabase.csv"
```

```
-- Column specification --------------------------------------------------
```

```
cols(
  isbn = col_double(),
  `e-isbn` = col_logical(),
  title = col_character(),
  auth_first = col_character(),
  auth_last = col_character(),
  trnsl_first = col_character(),
  trnsl_last = col_character(),
  publisher = col_character(),
  genre = col_character(),
  price = col_double(),
  pubdate_mo = col_character(),
  pubdate_yr = col_double(),
  language = col_character(),
  country = col_character(),
  addl_auth = col_character(),
  add_trnsl = col_character(),
  auth_gend = col_character(),
  trnsl_gend = col_character()
)
```

```
# translations= read_csv(file='translations.csv')
```

We should always familiarize ourselves with the dataset before beginning any kind of analysis.

```
class(translations)  # this tells us what kind of R object this is
```

```
[1] "spec_tbl_df" "tbl_df"      "tbl"         "data.frame"
```

```
dim(translations)  # 8628 rows by 18 columns
```

```
[1] 8628   18
```

```
head(translations)  # look at the first 6 (by default) rows of the dataset
```

```
# A tibble: 6 x 18
      isbn `e-isbn` title  auth_first auth_last trnsl_first trnsl_last
     <dbl> <lgl>    <chr>  <chr>      <chr>     <chr>       <chr>
1  9.78e12 NA       Equat~ Tom        Dreyer    Michiel     Heyns
2  9.78e12 NA       Blood~ Azem       Shkreli   Robert      Elsie
3  9.78e12 NA       Dream~ Valentina  Saracini  Erica       Weitzman
4  9.78e12 NA       Taxi   Khaled Al  Khamissi  Jonathan    Wright
5  9.79e12 NA       Final~ Abdelilah  Hamdouchi Jonathan    Smolin
6  9.79e12 NA       As Do~ Bahaa      Taher     Peter       Daniel
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

```r
tail(translations, n = 3)  # look at the last 3 rows of the dataset
```

```
# A tibble: 3 x 18
      isbn `e-isbn` title  auth_first auth_last trnsl_first trnsl_last
     <dbl> <lgl>    <chr>  <chr>      <chr>     <chr>       <chr>
1  9.78e12 NA       Orpha~ Serhiy     Zhadan    Reilly      Costigan-~
2  9.78e12 NA       Eleme~ Various    Various   Various     Various
3  9.78e12 NA       Begin~ Tran Le    Khanh     Bruce       Weigl
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

We can see that some observations have NA listed where no information is provided. We'll account for these missing values later on.

```r
# check the data type of each column in `translations`
str(translations, strict.width = "cut")
```

```
spec_tbl_df [8,628 x 18] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ isbn       : num [1:8628] 9.78e+12 9.78e+12 9.78e+12 9.78e+12 9.7..
 $ e-isbn     : logi [1:8628] NA NA NA NA NA NA ...
 $ title      : chr [1:8628] "Equatoria" "Blood of the Quill: Selec"..
 $ auth_first : chr [1:8628] "Tom" "Azem" "Valentina" "Khaled Al" ...
 $ auth_last  : chr [1:8628] "Dreyer" "Shkreli" "Saracini" "Khamiss"..
 $ trnsl_first: chr [1:8628] "Michiel" "Robert" "Erica" "Jonathan" ...
 $ trnsl_last : chr [1:8628] "Heyns" "Elsie" "Weitzman" "Wright" ...
 $ publisher  : chr [1:8628] "Aflame Books" "Green Integer" "Ugly D"..
 $ genre      : chr [1:8628] "Fiction" "Poetry" "Poetry" "Fiction" ...
 $ price      : num [1:8628] 15.9 13.9 15 15.9 17.9 ...
 $ pubdate_mo : chr [1:8628] "Oct" "Sep" "Jun" "Apr" ...
 $ pubdate_yr : num [1:8628] 2008 2008 2008 2008 2008 ...
 $ language   : chr [1:8628] "Afrikaans" "Albanian" "Albanian" "Ara"..
 $ country    : chr [1:8628] "South Africa" "Albania" "Albania" "Eg"..
 $ addl_auth  : chr [1:8628] NA NA NA NA ...
 $ add_trnsl  : chr [1:8628] NA "Janice Mathie-Heck;" "Rudina Jasin"..
 $ auth_gend  : chr [1:8628] "Male" "Male" "Female" "Male" ...
 $ trnsl_gend : chr [1:8628] "Male" "Both" "Female" "Male" ...
 - attr(*, "problems")= tibble [155 x 5] (S3: tbl_df/tbl/data.frame)
  ..$ row     : int [1:155] 1304 3706 3707 3813 3814 4446 4513 4514 ..
  ..$ col     : chr [1:155] "e-isbn" "e-isbn" "e-isbn" "e-isbn" ...
  ..$ expected: chr [1:155] "1/0/T/F/TRUE/FALSE" "1/0/T/F/TRUE/FALS"..
  ..$ actual  : chr [1:155] "9.78194E+12" "9.78194E+12" "9.78194E+1"..
  ..$ file    : chr [1:155] "'~/Documents/Princeton/CDH/SML 201/Mod"..
 - attr(*, "spec")=
  .. cols(
```

```
..   isbn = col_double(),
..   `e-isbn` = col_logical(),
..   title = col_character(),
..   auth_first = col_character(),
..   auth_last = col_character(),
..   trnsl_first = col_character(),
..   trnsl_last = col_character(),
..   publisher = col_character(),
..   genre = col_character(),
..   price = col_double(),
..   pubdate_mo = col_character(),
..   pubdate_yr = col_double(),
..   language = col_character(),
..   country = col_character(),
..   addl_auth = col_character(),
..   add_trnsl = col_character(),
..   auth_gend = col_character(),
..   trnsl_gend = col_character()
.. )
```

Notice that, next to each variable, the data type is listed. There are three basic data types in R that we'll use throughout this course: character, numeric, and logical.

(1) **Character variable**: each element is a string of one or more characters (essentially, letters)

(2) **Numeric variable**: each element is a number. The two most common classes of numeric variables are integer (whole number) and double (decimal)

(3) **Logical variable**: binary, two possible values (TRUE or FALSE).

Later on in the course, we'll also see factor variables, which are essentially numeric codes tied to character-valued levels.

Before analyzing the data, we should check each variable type in R. Here, we see that price is listed as a character variable, when it should in fact be a numeric variable. We can change this with a simple command. To reference a single column from the dataframe, we use the $ symbol.

```
translations$price <- as.numeric(translations$price)
```

Given that price is now numeric, we can find its corresponding 5-number summary:

```
summary(translations$price)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
    0.0    15.0    17.0    20.4    24.0   282.0      14
```

We can also input the entire dataset into the summary function, but in this case the only relevant output will be for price, given that the other variables are predominantly character

variables:

```
summary(translations)
```

```
      isbn            e-isbn           title
 Min.   :2.000e+00   Mode:logical   Length:8628
 1st Qu.:9.781e+12   NA's:8628      Class :character
 Median :9.782e+12                  Mode  :character
 Mean   :9.732e+12
 3rd Qu.:9.782e+12
 Max.   :9.870e+12
 NA's   :2
  auth_first         auth_last         trnsl_first
 Length:8628        Length:8628        Length:8628
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character
 trnsl_last         publisher            genre
 Length:8628        Length:8628        Length:8628
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character
     price        pubdate_mo          pubdate_yr     language
 Min.   :  0.0   Length:8628        Min.   :2008   Length:8628
 1st Qu.: 15.0   Class :character   1st Qu.:2012   Class :character
 Median : 17.0   Mode  :character   Median :2015   Mode  :character
 Mean   : 20.4                      Mean   :2015
 3rd Qu.: 24.0                      3rd Qu.:2018
 Max.   :282.0                      Max.   :2021
 NA's   :14
   country           addl_auth          add_trnsl
 Length:8628        Length:8628        Length:8628
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character
  auth_gend          trnsl_gend
 Length:8628        Length:8628
 Class :character   Class :character
 Mode  :character   Mode  :character
```

Manipulating Variables in a Data Frame with dplyr

The dpylr package in R has several useful functions to manuipulate a data frame. Here we'll use the translation database to try out these functions.

Using select( )

Let's have another look at an overview of our dataset:

```r
# check the data type of each column in `translations`
str(translations, strict.width = "cut")
```

```
spec_tbl_df [8,628 x 18] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ isbn       : num [1:8628] 9.78e+12 9.78e+12 9.78e+12 9.78e+12 9.7..
 $ e-isbn     : logi [1:8628] NA NA NA NA NA NA ...
 $ title      : chr [1:8628] "Equatoria" "Blood of the Quill: Selec"..
 $ auth_first : chr [1:8628] "Tom" "Azem" "Valentina" "Khaled Al" ...
 $ auth_last  : chr [1:8628] "Dreyer" "Shkreli" "Saracini" "Khamiss"..
 $ trnsl_first: chr [1:8628] "Michiel" "Robert" "Erica" "Jonathan" ...
 $ trnsl_last : chr [1:8628] "Heyns" "Elsie" "Weitzman" "Wright" ...
 $ publisher  : chr [1:8628] "Aflame Books" "Green Integer" "Ugly D"..
 $ genre      : chr [1:8628] "Fiction" "Poetry" "Poetry" "Fiction" ...
 $ price      : num [1:8628] 15.9 13.9 15 15.9 17.9 ...
 $ pubdate_mo : chr [1:8628] "Oct" "Sep" "Jun" "Apr" ...
 $ pubdate_yr : num [1:8628] 2008 2008 2008 2008 2008 ...
 $ language   : chr [1:8628] "Afrikaans" "Albanian" "Albanian" "Ara"..
 $ country    : chr [1:8628] "South Africa" "Albania" "Albania" "Eg"..
 $ addl_auth  : chr [1:8628] NA NA NA NA ...
 $ add_trnsl  : chr [1:8628] NA "Janice Mathie-Heck;" "Rudina Jasin"..
 $ auth_gend  : chr [1:8628] "Male" "Male" "Female" "Male" ...
 $ trnsl_gend : chr [1:8628] "Male" "Both" "Female" "Male" ...
 - attr(*, "problems")= tibble [155 x 5] (S3: tbl_df/tbl/data.frame)
  ..$ row     : int [1:155] 1304 3706 3707 3813 3814 4446 4513 4514 ..
  ..$ col     : chr [1:155] "e-isbn" "e-isbn" "e-isbn" "e-isbn" ...
  ..$ expected: chr [1:155] "1/0/T/F/TRUE/FALSE" "1/0/T/F/TRUE/FALS"..
  ..$ actual  : chr [1:155] "9.78194E+12" "9.78194E+12" "9.78194E+1"..
  ..$ file    : chr [1:155] "'~/Documents/Princeton/CDH/SML 201/Mod"..
 - attr(*, "spec")=
  .. cols(
  ..   isbn = col_double(),
  ..   `e-isbn` = col_logical(),
  ..   title = col_character(),
  ..   auth_first = col_character(),
```

```
  ..   auth_last = col_character(),
  ..   trnsl_first = col_character(),
  ..   trnsl_last = col_character(),
  ..   publisher = col_character(),
  ..   genre = col_character(),
  ..   price = col_double(),
  ..   pubdate_mo = col_character(),
  ..   pubdate_yr = col_double(),
  ..   language = col_character(),
  ..   country = col_character(),
  ..   addl_auth = col_character(),
  ..   add_trnsl = col_character(),
  ..   auth_gend = col_character(),
  ..   trnsl_gend = col_character()
  .. )
```

There are several variables that might be less useful to us in analyzing the data. ISBN and e-ISBN numbers, for instance, might not tell us much. We can keep certain variables (columns) while removing others by using the `select( )` function in the dplyr package. For instance:

```
translations %>%
    select(title, genre, country, language, price)  #create data frame with the selected variabl
```

```
# A tibble: 8,628 x 5
   title                              genre   country    language price
   <chr>                              <chr>   <chr>      <chr>    <dbl>
 1 Equatoria                          Fiction South Afr~ Afrikaa~  16.0
 2 Blood of the Quill: Selected Poe~ Poetry  Albania    Albanian  14.0
 3 Dreaming Escape                    Poetry  Albania    Albanian  15
 4 Taxi                               Fiction Egypt      Arabic    16.0
 5 Final Bet                          Fiction Morocco    Arabic    18.0
 6 As Doha Said                       Fiction Egypt      Arabic    23.0
 7 Final Night                        Fiction Iraq       Arabic    18.0
 8 Cell Block Five                    Fiction Iraq       Arabic    18.0
 9 Crane                              Fiction Syria      Arabic    19.0
10 Gold Dust                          Fiction Libya      Arabic    18.0
# ... with 8,618 more rows
```

This new dataframe, with fewer variables, will be easier to work with. Essentially we're tuning out the extra noise. We can assign it a new name:

```
new_translations <- translations %>%
    select(title, genre, country, language, price)
```

Alternatively, we might want to select columns (variables) according to their position in the dataframe.

```
translations %>%
    select(3:5)  #select columns 3-5
```

```
# A tibble: 8,628 x 3
   title                                          auth_first auth_last
   <chr>                                          <chr>      <chr>
 1 Equatoria                                      Tom        Dreyer
 2 Blood of the Quill: Selected Poetry from Koso~ Azem       Shkreli
 3 Dreaming Escape                                Valentina  Saracini
 4 Taxi                                           Khaled Al  Khamissi
 5 Final Bet                                      Abdelilah  Hamdouchi
 6 As Doha Said                                   Bahaa      Taher
 7 Final Night                                    Buthaina ~ Nasiri
 8 Cell Block Five                                Fadhil Al  Azzawi
 9 Crane                                          Halim      Barakat
10 Gold Dust                                      Ibrahim Al Koni
# ... with 8,618 more rows
```

```
translations %>%
    select(3:ncol(translations))  #select all columns beginning with the 3rd column
```

```
# A tibble: 8,628 x 16
   title   auth_first auth_last trnsl_first trnsl_last publisher genre
   <chr>   <chr>      <chr>     <chr>       <chr>      <chr>     <chr>
 1 Equato~ Tom        Dreyer    Michiel     Heyns      Aflame B~ Fict~
 2 Blood ~ Azem       Shkreli   Robert      Elsie      Green In~ Poet~
 3 Dreami~ Valentina  Saracini  Erica       Weitzman   Ugly Duc~ Poet~
 4 Taxi    Khaled Al  Khamissi  Jonathan    Wright     Aflame B~ Fict~
 5 Final ~ Abdelilah  Hamdouchi Jonathan    Smolin     American~ Fict~
 6 As Doh~ Bahaa      Taher     Peter       Daniel     American~ Fict~
 7 Final ~ Buthaina ~ Nasiri    Denys       Johnson-D~ American~ Fict~
 8 Cell B~ Fadhil Al  Azzawi    William M.  Hutchins   American~ Fict~
 9 Crane   Halim      Barakat   Bassam      Frangieh   American~ Fict~
10 Gold D~ Ibrahim Al Koni      Elliott     Colla      American~ Fict~
# ... with 8,618 more rows, and 9 more variables: price <dbl>,
#   pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

```
translations %>%
    select(, -1)  #remove only the first column
```

```
# A tibble: 8,628 x 17
   `e-isbn` title          auth_first auth_last trnsl_first trnsl_last
   <lgl>    <chr>          <chr>      <chr>     <chr>       <chr>
 1 NA       Equatoria      Tom        Dreyer    Michiel     Heyns
 2 NA       Blood of the ~ Azem       Shkreli   Robert      Elsie
 3 NA       Dreaming Esca~ Valentina  Saracini  Erica       Weitzman
 4 NA       Taxi           Khaled Al  Khamissi  Jonathan    Wright
 5 NA       Final Bet      Abdelilah  Hamdouchi Jonathan    Smolin
 6 NA       As Doha Said   Bahaa      Taher     Peter       Daniel
 7 NA       Final Night    Buthaina ~ Nasiri    Denys       Johnson-D~
```

```
 8 NA       Cell Block Fi~ Fadhil Al  Azzawi    William M.  Hutchins
 9 NA       Crane          Halim      Barakat   Bassam      Frangieh
10 NA       Gold Dust      Ibrahim Al Koni      Elliott     Colla
# ... with 8,618 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

Using mutate( )

In some cases, we may want to change a column or add an additional column to the dataframe. Suppose that we are presenting to an international audience and would like to include the book prices in dollars and euros. We can add a price column in euros using the `mutate( )` function. (Note: at the moment, 1USD is approximately equal to 0.85 euros)

```
translations %>%
    mutate(EuroPrice = price * 0.85)
```

```
# A tibble: 8,628 x 19
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Equa~ Tom        Dreyer    Michiel     Heyns
 2  9.78e12 NA       Bloo~ Azem       Shkreli   Robert      Elsie
 3  9.78e12 NA       Drea~ Valentina  Saracini  Erica       Weitzman
 4  9.78e12 NA       Taxi  Khaled Al  Khamissi  Jonathan    Wright
 5  9.79e12 NA       Fina~ Abdelilah  Hamdouchi Jonathan    Smolin
 6  9.79e12 NA       As D~ Bahaa      Taher     Peter       Daniel
 7  9.79e12 NA       Fina~ Buthaina ~ Nasiri    Denys       Johnson-D~
 8  9.79e12 NA       Cell~ Fadhil Al  Azzawi    William M.  Hutchins
 9  9.79e12 NA       Crane Halim      Barakat   Bassam      Frangieh
10  9.79e12 NA       Gold~ Ibrahim Al Koni      Elliott     Colla
# ... with 8,618 more rows, and 12 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>, EuroPrice <dbl>
```

Note that the new column is given the name "EuroPrice." We could have written "price=price*0.85" within our function, but this would have overwritten the original column in the dataset. We recommend against altering the original dataset, that way you can always return to the original data.

Using pull( )

If you want to extract the values in a column as a vector, you can use the `pull( )` function. Suppose we want to make a list of all of the countries included in the translation database.

```
translations %>% pull(country) %>% #extract vector with country names
    unique() %>% #list each country only once
    sort() #list in alphabetical order
```

```
 [1] "----------------"
 [2] "Afghanistan"
 [3] "Albania"
 [4] "Algeria"
 [5] "Angola"
 [6] "Argentina"
 [7] "Armenia"
 [8] "Australia"
 [9] "Austria"
[10] "Azerbaijan"
[11] "Bahrain"
[12] "Bangladesh"
[13] "Belarus"
[14] "Belgium"
[15] "Benin"
[16] "Bolivia"
[17] "Bosnia-Herzegovina"
[18] "Brazil"
[19] "Bulgaria"
[20] "Burundi"
[21] "Cambodia"
[22] "Cameroon"
[23] "Canada"
[24] "Cape Verde"
[25] "Chile"
[26] "China"
[27] "Colombia"
[28] "Costa Rica"
[29] "Cote D'Ivoire"
[30] "Croatia"
[31] "Cuba"
[32] "Cyprus"
[33] "Czech Republic"
[34] "Democratic Peoples Republic of Korea"
[35] "Democratic Republic of the Congo"
[36] "Denmark"
[37] "Djibouti"
[38] "Dominican Republic"
[39] "Ecuador"
[40] "Egypt"
[41] "El Salvador"
[42] "Equatorial Guinea"
[43] "Eritrea"
[44] "Estonia"
[45] "Ethiopia"
[46] "Faroe Islands"
[47] "Finland"
[48] "France"
```

```
[49] "Gabon"
[50] "Georgia"
[51] "Germany"
[52] "Greece"
[53] "Greenland"
[54] "Guadeloupe"
[55] "Guatemala"
[56] "Guinea"
[57] "Haiti"
[58] "Honduras"
[59] "Hong Kong"
[60] "Hungary"
[61] "Iceland"
[62] "India"
[63] "Indonesia"
[64] "Iran"
[65] "Iraq"
[66] "Ireland"
[67] "Israel"
[68] "Italy"
[69] "Japan"
[70] "Jordan"
[71] "Kazakhstan"
[72] "Kenya"
[73] "Korea"
[74] "Kuwait"
[75] "Kyrgyzstan"
[76] "Latvia"
[77] "Lebanon"
[78] "Libya"
[79] "Liechtenstein"
[80] "Lithuania"
[81] "Luxembourg"
[82] "Macedonia"
[83] "Madagascar"
[84] "Malaysia"
[85] "Mali"
[86] "Martinique"
[87] "Mauritius"
[88] "Mexico"
[89] "Moldova"
[90] "Mongolia"
[91] "Montenegro"
[92] "Morocco"
[93] "Mozambique"
[94] "Myanmar"
[95] "Netherlands"
[96] "Netherlands Antilles"
```

```
 [97] "Nicaragua"
 [98] "Niger"
 [99] "Norway"
[100] "Oman"
[101] "Pakistan"
[102] "Palestine"
[103] "Peru"
[104] "Poland"
[105] "Portugal"
[106] "Puerto Rico"
[107] "Qatar"
[108] "Republic of the Congo"
[109] "Romania"
[110] "Russia"
[111] "Rwanda"
[112] "Saint Martin"
[113] "Saudi Arabia"
[114] "Senegal"
[115] "Serbia"
[116] "Serbia and Montenegro"
[117] "Singapore"
[118] "Slovakia"
[119] "Slovenia"
[120] "Somalia"
[121] "South Africa"
[122] "Spain"
[123] "Sudan"
[124] "Sweden"
[125] "Switzerland"
[126] "Syria"
[127] "Taiwan"
[128] "Tanzania"
[129] "Thailand"
[130] "Togo"
[131] "Tunisia"
[132] "Turkey"
[133] "Turkmenistan"
[134] "Ukraine"
[135] "United Arab Emirates"
[136] "United Kingdom"
[137] "United States"
[138] "Uruguay"
[139] "Uzbekistan"
[140] "Venezuela"
[141] "Vietnam"
[142] "Western Sahara"
[143] "Yemen"
```

That first entry with the dashes might seem a bit strange. If we go back to the original dataset, we'll see that this corresponds to some translations that include work from multiple countries. These exceptional cases are important to fully understanding the dataset, so make sure to spend some time with them.

Manipulating Cases in a Data Frame in dplyr

Note that functions such as `select( )` and `mutate( )` have to do with the variables (columns) of the data frame. Sometimes, we may want to extract particular observations (rows) from the data frame, without altering the variables.

Using filter( )

To extract rows that meet certain logical criteria, we can use the `filter( )` function in dplyr. Suppose that our area of research interest is Spanish, and we only want to look at translations from Spanish.

```
translations %>%
    filter(language == "Spanish")
```

```
# A tibble: 1,171 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Port~ Jaime Luis Huenun    Daniel      Borzutzky
 2  9.78e12 NA       Fear~ Enrique    Serna     Georgina    Jimenez
 3  9.78e12 NA       Cade~ Oscar Nun~ Olivas    Joanna      Griffin
 4  9.78e12 NA       Fina~ Ronald     Flores    Gavin       O'Toole
 5  9.78e12 NA       Wate~ Domingo    Villar    Martin      Schifino
 6  9.78e12 NA       Unde~ Rosario    Sanmiguel John        Pluecker
 7  9.78e12 NA       Mule  Juan Esla~ Galan     Lisa        Dillman
 8  9.78e12 NA       Hava~ Leonardo   Padura    Peter       Bush
 9  9.78e12 NA       Quix~ Ramon      Fernandez Alfred      Noyes
10  9.78e12 NA       Stat~ Cristina ~ Rossi     Marilyn     Buck
# ... with 1,161 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

It's easy to add multiple criteria. Suppose we're interested only in Fiction translated from Spanish.

```
translations %>%
    filter(language == "Spanish" & genre == "Fiction")  #using the AND logical operator
```

```
# A tibble: 725 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Fear~ Enrique    Serna     Georgina    Jimenez
```

```
 2  9.78e12 NA       Cade~ Oscar Nun~ Olivas    Joanna      Griffin
 3  9.78e12 NA       Fina~ Ronald     Flores    Gavin       O'Toole
 4  9.78e12 NA       Wate~ Domingo    Villar    Martin      Schifino
 5  9.78e12 NA       Unde~ Rosario    Sanmiguel John        Pluecker
 6  9.78e12 NA       Mule  Juan Esla~ Galan     Lisa        Dillman
 7  9.78e12 NA       Hava~ Leonardo   Padura    Peter       Bush
 8  9.78e12 NA       Marg~ Sergio     Ramirez   Michael     Miller
 9  9.78e12 NA       Hele~ Julian     Ayesta    Margaret J~ Costa
10  9.78e12 NA       Deat~ Alicia     Gimenez-~ Jonathan    Dunne
# ... with 715 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

Or maybe we're interested in two particular countries, Mexico and Nicaragua.

```
translations %>%
    filter(country == "Mexico" | country == "Nicaragua")  #using the OR logical operator
```

```
# A tibble: 202 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Fear~ Enrique    Serna     Georgina    Jimenez
 2  9.78e12 NA       Unde~ Rosario    Sanmiguel John        Pluecker
 3  9.78e12 NA       Befo~ David      Huerta    Mark        Schafer
 4  9.78e12 NA       Marg~ Sergio     Ramirez   Michael     Miller
 5  9.79e12 NA       Sun,~ Jorge      Hernandez various     various
 6  9.78e12 NA       Curs~ Liliana    Blum      Toshiya     Kamei
 7  9.78e12 NA       Look~ Miguel     Gonzalez~ Miguel      Gonzalez-~
 8  9.78e12 NA       Happ~ Carlos     Fuentes   Edith       Grossman
 9  9.78e12 NA       Sele~ Victor Ma~ Mendiola  Ruth        Fainlight
10  9.78e12 NA       Tues~ Cecilia    Urbina    Clare       Sullivan
# ... with 192 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

Or in translations from Spanish that fall in the price range of 10-20 dollars.

```
translations %>%
    filter(language == "Spanish" & price >= 10 & price <= 20)  #GREATER THAN OR EQUAL TO and LES
```

```
# A tibble: 874 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Port~ Jaime Luis Huenun    Daniel      Borzutzky
 2  9.78e12 NA       Fear~ Enrique    Serna     Georgina    Jimenez
 3  9.78e12 NA       Cade~ Oscar Nun~ Olivas    Joanna      Griffin
 4  9.78e12 NA       Fina~ Ronald     Flores    Gavin       O'Toole
 5  9.78e12 NA       Wate~ Domingo    Villar    Martin      Schifino
```

```
 6  9.78e12 NA       Unde~ Rosario    Sanmiguel John        Pluecker
 7  9.78e12 NA       Mule  Juan Esla~ Galan     Lisa        Dillman
 8  9.78e12 NA       Hava~ Leonardo   Padura    Peter       Bush
 9  9.78e12 NA       Quix~ Ramon      Fernandez Alfred      Noyes
10  9.78e12 NA       Stat~ Cristina ~ Rossi     Marilyn     Buck
# ... with 864 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

Or maybe we want to exclude a country: translations from Spanish, not including Spain.

```
translations %>%
    filter(language == "Spanish" & country != "Spain")  #making an exclusion with NOT
```

```
# A tibble: 840 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Port~ Jaime Luis Huenun    Daniel      Borzutzky
 2  9.78e12 NA       Fear~ Enrique    Serna     Georgina    Jimenez
 3  9.78e12 NA       Cade~ Oscar Nun~ Olivas    Joanna      Griffin
 4  9.78e12 NA       Fina~ Ronald     Flores    Gavin       O'Toole
 5  9.78e12 NA       Unde~ Rosario    Sanmiguel John        Pluecker
 6  9.78e12 NA       Hava~ Leonardo   Padura    Peter       Bush
 7  9.78e12 NA       Stat~ Cristina ~ Rossi     Marilyn     Buck
 8  9.78e12 NA       Befo~ David      Huerta    Mark        Schafer
 9  9.78e12 NA       Hand~ Pablo      Neruda    William     O'Daly
10  9.78e12 NA       Worl~ Pablo      Neruda    William     O'Daly
# ... with 830 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

The `filter()` function is a useful way to limit the observations to only those of interest. It will also come in handy later on when we want to graph observations that meet certain criteria.

Using slice( ) and related functions

To select rows by position, use the `slice( )` function.

```
translations %>%
    slice(5:10)  #selects rows 5-10
```

```
# A tibble: 6 x 18
      isbn `e-isbn` title  auth_first auth_last trnsl_first trnsl_last
     <dbl> <lgl>    <chr>  <chr>      <chr>     <chr>       <chr>
1  9.79e12 NA       Final~ Abdelilah  Hamdouchi Jonathan    Smolin
2  9.79e12 NA       As Do~ Bahaa      Taher     Peter       Daniel
3  9.79e12 NA       Final~ Buthaina ~ Nasiri    Denys       Johnson-D~
4  9.79e12 NA       Cell ~ Fadhil Al  Azzawi    William M.  Hutchins
```

```
5  9.79e12 NA       Crane  Halim      Barakat   Bassam      Frangieh
6  9.79e12 NA       Gold ~ Ibrahim Al Koni      Elliott     Colla
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

To select rows with the lowest values, use `slice_min( )`.

```
translations %>%
    slice_min(price)  #there are 20 translations at the minimum cost of 0 dollars
```

```
# A tibble: 20 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Same~ Jaan       Kaplinski Susan       Wilson
 2  9.78e12 NA       Anti~ Joseph     Roth      Richard     Panchyk
 3  9.78e12 NA       Book~ Ilhan      Berk      George      Messo
 4  9.79e12 NA       Vert~ Ahmed      Mourad    Robin       Moger
 5  9.79e12 NA       Amer~ Inaam      Kachachi  Nariman     Youssef
 6  9.79e12 NA       Toba~ Ali        Bader     Amira       Nowaira
 7  9.79e12 NA       Iraq~ Samuel     Shimon    Christina   Phillips
 8  9.78e12 NA       His ~ Ye         Mimi      Steve       Bradbury
 9  9.78e12 NA       Tahr~ Various    Various   Maged       Zaher
10  9.79e12 NA       Subl~ Yuri       Mamleyev  Marian      Schwartz
11  9.78e12 NA       All ~ Walter     Kempowski Anthea      Bell
12  9.78e12 NA       Off ~ Inge       Merkel    Dennis      McCort
13  9.78e12 NA       Pari~ Wolfgang   Hermann   Mark        Miscovich
14  9.78e12 NA       Writ~ Various    Various   Various     Various
15  9.78e12 NA       Pano~ Dusan      Sarotar   Rawley      Grau
16  9.78e12 NA       Othe~ Maria Eug~ Vaz Ferr~ William F.  Blair
17  9.78e12 NA       Moni~ Florencia  Castella~ Alexis      Almeida
18  9.78e12 NA       What~ Vladimir   Aristov   Julia       Trubikhina
19  9.78e12 NA       Book~ Claudio    Romo      David       Haughton
20  9.78e12 NA       Jour~ Claudio    Romo      David       Haughton
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

```
translations %>%
    slice_min(price, prop = 0.25)  #returns the 25% of all translations with the lowest cost
```

```
# A tibble: 2,527 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Same~ Jaan       Kaplinski Susan       Wilson
 2  9.78e12 NA       Anti~ Joseph     Roth      Richard     Panchyk
 3  9.78e12 NA       Book~ Ilhan      Berk      George      Messo
 4  9.79e12 NA       Vert~ Ahmed      Mourad    Robin       Moger
```

```
 5  9.79e12 NA       Amer~ Inaam      Kachachi  Nariman     Youssef
 6  9.79e12 NA       Toba~ Ali        Bader     Amira       Nowaira
 7  9.79e12 NA       Iraq~ Samuel     Shimon    Christina   Phillips
 8  9.78e12 NA       His ~ Ye         Mimi      Steve       Bradbury
 9  9.78e12 NA       Tahr~ Various    Various   Maged       Zaher
10  9.79e12 NA       Subl~ Yuri       Mamleyev  Marian      Schwartz
# ... with 2,517 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

To select rows with the highest values, use `slice_max( )`.

```
translations %>%
    slice_max(price)  #there is 1 translation at the maximum cost of 282 dollars
```

```
# A tibble: 1 x 18
      isbn `e-isbn` title  auth_first auth_last trnsl_first trnsl_last
     <dbl> <lgl>    <chr>  <chr>      <chr>     <chr>       <chr>
1  9.79e12 NA       Chris~ Hans-Jurg~ Prien     Stephen     Buckwalter
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

```
translations %>%
    slice_max(price, prop = 0.1)  #returns the 10% of all translations with the highest cost
```

```
# A tibble: 881 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.79e12 NA       Chri~ Hans-Jurg~ Prien     Stephen     Buckwalter
 2  9.78e12 NA       Pyth~ Leonid     Zhmud     Kevin       Windle
 3  9.79e12 NA       Unwe~ Daniela    Gleizer   Susan       Thomae
 4  9.79e12 NA       Euro~ Friedrich  Lenger    Joel        Golb
 5  9.79e12 NA       Cult~ Mingyuan   Gu        Wang        Juefei
 6  9.78e12 NA       The ~ Hanne Kol~ Poulsen   Hanne       Hammer
 7  9.79e12 NA       War ~ Pelai      Pages i ~ Patrick     Gallagher
 8  9.79e12 NA       Sola~ Gabriele   Zini      Pei-Shu     Wu
 9  9.78e12 NA       Gran~ Christian  Heck      John        Goodman
10  9.79e12 NA       Ever~ Organizat~ Korean H~ Michael     Shin
# ... with 871 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

To randomly select a subset of rows, use `slice_sample( )`.

```
translations %>%
    slice_sample(n = 10)  #randomly select 10 rows
```

```
# A tibble: 10 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Visi~ Miguel An~ Bustos    Lucina      Schell
 2  9.78e12 NA       All ~ Dorit      Rabinyan  Jessica     Cohen
 3  9.78e12 NA       Unra~ Bernard    Fernandez Georges     Ripka
 4  9.78e12 NA       Ridd~ Lin        Zhe       George      Fowler
 5  9.78e12 NA       Beau~ Marcel     Ayme      Sophie      Lewis
 6  9.78e12 NA       Taxi  Khaled Al  Khamissi  Jonathan    Wright
 7  9.78e12 NA       Brie~ Enrique    Vila-Mat~ Tom         Bunstead
 8  9.79e12 NA       Defe~ Jiri       Priban    Stuart      Hoskins
 9  9.78e12 NA       Rave~ Gabi       Kreslehn~ Alison      Layland
10  9.78e12 NA       Stoc~ Nathan     Larson    Various     Various
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

Note that if you run `slice_sample( )` multiple times, you should get a new random selection each time.

If you want to create a simulation that can be reproduced, use the `set.seed( )` function.

```r
set.seed(800)  #you can set the seed with any number
translations %>%
    slice_sample(n = 10)
```

```
# A tibble: 10 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Bloo~ Sebastia   Alzamora  Maruxa      Relano
 2  9.78e12 NA       Ova ~ Susana     Thenon    Rebekah     Smith
 3  9.78e12 NA       Mong~ Rafael     Bernal    Katherine   Silver
 4  9.78e12 NA       Paul~ Anzhelina  Polonska~ Andrew      Wachtel
 5  9.78e12 NA       Die,~ Ariana     Harwicz   Sara        Moses
 6  9.78e12 NA       Germ~ Mark Osbo~ Humphries Wilhelm     Kiesselba~
 7  9.78e12 NA       Jour~ Radwa      Ashour    Michelle    Hartman
 8  9.78e12 NA       Thre~ Lianke     Yan       Carlos      Rojas
 9  9.78e12 NA       Rain~ Andrea     Hirata    Angie       Kilbane
10  9.78e12 NA       Devi~ Jo         Nesbo     Don         Bartlett
# ... with 11 more variables: publisher <chr>, genre <chr>,
#   price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>, language <chr>,
#   country <chr>, addl_auth <chr>, add_trnsl <chr>, auth_gend <chr>,
#   trnsl_gend <chr>
```

Every time you run the chunk above, you will get the same randomly selected 10 rows.

Using arrange( )

Sometimes we may want to rearrange the order of our dataset. Let's `arrange( )` the dataset by price:

```
translations %>%
    arrange(price)  # orders from lowest price to highest price
```

```
# A tibble: 8,628 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Same~ Jaan       Kaplinski Susan       Wilson
 2  9.78e12 NA       Anti~ Joseph     Roth      Richard     Panchyk
 3  9.78e12 NA       Book~ Ilhan      Berk      George      Messo
 4  9.79e12 NA       Vert~ Ahmed      Mourad    Robin       Moger
 5  9.79e12 NA       Amer~ Inaam      Kachachi  Nariman     Youssef
 6  9.79e12 NA       Toba~ Ali        Bader     Amira       Nowaira
 7  9.79e12 NA       Iraq~ Samuel     Shimon    Christina   Phillips
 8  9.78e12 NA       His ~ Ye         Mimi      Steve       Bradbury
 9  9.78e12 NA       Tahr~ Various    Various   Maged       Zaher
10  9.79e12 NA       Subl~ Yuri       Mamleyev  Marian      Schwartz
# ... with 8,618 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

```
translations %>%
    arrange(desc(price))  # orders from highest price to lowest price
```

```
# A tibble: 8,628 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.79e12 NA       Chri~ Hans-Jurg~ Prien     Stephen     Buckwalter
 2  9.78e12 NA       Pyth~ Leonid     Zhmud     Kevin       Windle
 3  9.79e12 NA       Unwe~ Daniela    Gleizer   Susan       Thomae
 4  9.79e12 NA       Euro~ Friedrich  Lenger    Joel        Golb
 5  9.79e12 NA       Cult~ Mingyuan   Gu        Wang        Juefei
 6  9.78e12 NA       The ~ Hanne Kol~ Poulsen   Hanne       Hammer
 7  9.79e12 NA       War ~ Pelai      Pages i ~ Patrick     Gallagher
 8  9.79e12 NA       Sola~ Gabriele   Zini      Pei-Shu     Wu
 9  9.78e12 NA       Gran~ Christian  Heck      John        Goodman
10  9.79e12 NA       Ever~ Organizat~ Korean H~ Michael     Shin
# ... with 8,618 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

Using group_by( ) and summarize( )

Sometimes we may want to create certain groups within the dataset to carry out our analysis. Suppose we want to separate out the mean (average) price for books by country.

```
translations %>%
    group_by(country) %>%
    summarize(AveragePrice = mean(price))
```

```
# A tibble: 144 x 2
   country         AveragePrice
   <chr>                  <dbl>
 1 ---------------         19.6
 2 Afghanistan             18.5
 3 Albania                 18.7
 4 Algeria                 17.8
 5 Angola                  15.7
 6 Argentina               17.4
 7 Armenia                 19.9
 8 Australia               38.3
 9 Austria                 20.3
10 Azerbaijan              24.8
# ... with 134 more rows
```

We can also `group_by` more than one variable. R will group from left to right. Can you tell the difference between the following two results?

```
translations %>% group_by(genre, country) %>% #groups by genre and then country
    summarize(AveragePrice=mean(price))
```

```
`summarise()` has grouped output by 'genre'. You can override using the `.groups` argument.
```

```
# A tibble: 348 x 3
# Groups:   genre [4]
   genre      country     AveragePrice
   <chr>      <chr>              <dbl>
 1 Children's Argentina           18.7
 2 Children's Austria             18.0
 3 Children's Belgium             17.2
 4 Children's Brazil              19.4
 5 Children's Canada              18.5
 6 Children's Chile               18.2
 7 Children's China               13.7
 8 Children's Colombia            16.5
 9 Children's Denmark             10.8
10 Children's El Salvador         19.0
# ... with 338 more rows
```

```
translations %>% group_by(country, genre) %>% #groups by country and then genre
    summarize(AveragePrice=mean(price))
```

```
`summarise()` has grouped output by 'country'. You can override using the `.groups` argument.
```

```
# A tibble: 348 x 3
# Groups:   country [144]
   country         genre      AveragePrice
   <chr>           <chr>             <dbl>
 1 --------------- Fiction            21.0
 2 --------------- Poetry             17.0
 3 Afghanistan     Fiction            16.0
 4 Afghanistan     Poetry             21.8
 5 Albania         Fiction            21.0
 6 Albania         Nonfiction         18.5
 7 Albania         Poetry             15.0
 8 Algeria         Fiction            17.5
 9 Algeria         Nonfiction         22.0
10 Algeria         Poetry             17.6
# ... with 338 more rows
```

Note that the `group_by` function does not alter the original dataframe. It simply creates groups within the data to facilitate analysis.

Using count( ) and n( )

To count the number of rows (observations) in a particular group, we can use the `count( )` function.

```
translations %>%
    count(country)  # counts the number of translations by country
```

```
# A tibble: 144 x 2
   country             n
   <chr>           <int>
 1 ---------------     3
 2 Afghanistan         7
 3 Albania            18
 4 Algeria            34
 5 Angola              7
 6 Argentina         247
 7 Armenia            12
 8 Australia           3
 9 Austria           117
10 Azerbaijan          3
# ... with 134 more rows
```

To count the number of rows within the summarize function, use the `n( )` function. For instance, we can obtain the same result as above with the following code:

```
translations %>%
    group_by(country) %>%
    summarize(n())
```

```
# A tibble: 144 x 2
   country         `n()`
```

```
   <chr>           <int>
 1 ---------------     3
 2 Afghanistan         7
 3 Albania            18
 4 Algeria            34
 5 Angola              7
 6 Argentina         247
 7 Armenia            12
 8 Australia           3
 9 Austria           117
10 Azerbaijan          3
# ... with 134 more rows
```

To count the number of uniques, use the `n_distinct( )` function.

```
translations %>%
    summarize(n_distinct(country))  # determine the number of distinct countries
```

```
# A tibble: 1 x 1
  `n_distinct(country)`
                  <int>
1                   144
```

These are some of the many dplyr functions available to you. For a cheat sheet that includes these and more, see https://github.com/rstudio/cheatsheets/blob/master/data-transformation.pdf.

Summary functions for numeric variables

We will now turn our attention to numeric variables, which we'll analyze in terms of center and spread. One numeric variable in the translation database is `price`.

Three ways to summarize the central location and the spread of a dataset:

- **Mean, standard deviation/variance**
- **Five number summary: minimum, 1st quartile, median(i.e., 2nd quartile), 3rd quartile, maximum**
- **Median, interquartile range**

Mean, standard deviation and variance

Recall that the **mean** of a set of data points is the *average* of all the data points in the set

$$mean_of_a_dataset = \frac{pt.1 + pt.2 + ... + pt.n}{n}$$

$pt.i$ stands for the ith data point in your dataset

Variance gives a measure of the spread of the dataset by taking the average of the squared difference between each data point and the mean of the dataset:

$$variance_of_dataset \tag{1}$$
$$= \frac{(pt.1 - mean)^2 + (pt.2 - mean)^2 + ... + (pt.n - mean)^2}{n - 1} \tag{2}$$

A question for you: Why do we need to square the difference?

Note: the mean of a dataset is the quantity that minimizes the average of the squared difference between the data points and the quantity.

Taking square root of the variance gives standard deviation an unit that is consistent with the units for the original dataset.

$$SD_of_dataset = \sqrt{variance_of_dataset}$$

Five number summary: minimum, 1st quartile, median, 3rd quartile, maximum

---

Recall that 1st quartile, median and 3rd quartile are just 25th , 50th and 75th percentiles, respectively.

While mean and SD give a quick summary of the central location and the spread of a dataset, they do not provide information on the shape of the distribution; e.g., we cannot tell from the values of mean and SD whether the distribution is symmetric or if there are extreme values in the dataset.

The following three datasets have their means approximately 0 and their SDs approximately 1. However, we can see that the shapes of their distributions are quite different. The first two distributions are symmetric while the last one is skewed right (i.e., with a long right tail).

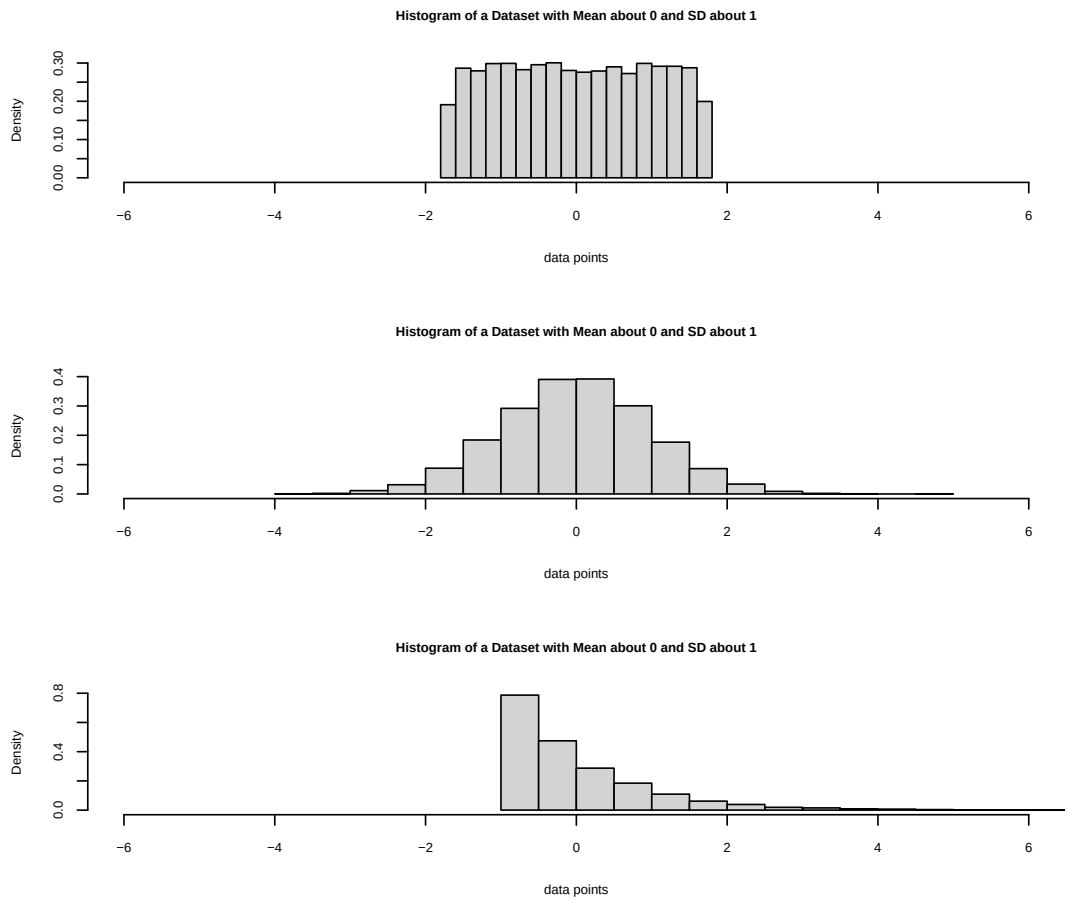


Figure 1: Histograms for three datasets each with mean around 0 and SD around 1

The five number summary gives a better idea of the shape of the distribution of a dataset: The median gives an idea of the central location (different definition of central location from the definition for the mean). For the three datasets above the five number summary along with the means are the following accordingly:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-1.73	-0.87	-0.01	0	0.88	1.73
-3.77	-0.67	0.00	0	0.66	4.59
-1.00	-0.71	-0.31	0	0.41	7.48

Compared to the median, the mean was affected by the extreme values in the third dataset.

Median, interquartile range

We already know that median is just the 50th percentile.

Interquartile range (IQR) gives a measure of the spread of the "central" 50% of the data: Interquartile range = Q3-Q1

Finding measures of center and spread

Let's try to find the mean (average) price of a translation in the translation database. Recall that to refer to a particular column, we use the $ symbol.

```
mean(translations$price)  # notice that this results in an error
```

```
[1] NA
```

An error is often the result of missing values in the dataset. There are several translations in the translation database that do not have a price listed. We can remove these entries.

```
mean(translations$price, na.rm = TRUE)
```

```
[1] 20.40069
```

We should be careful about excluding too much data. Let's check to see how many entries were excluded.

```
sum(is.na(translations$price))  # count how many entries do not have a price listed
```

```
[1] 14
```

We might also recall that the 5-number summary includes the number of NAs.

```
summary(translations$price)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
    0.0    15.0    17.0    20.4    24.0   282.0      14 
```

This small number of exclusions shouldn't have a major effect on our results.

Standard deviation and variance are common measures of spread, which we'll study in depth later on in the course.

```
sd(translations$price, na.rm = T)  # calculate the standard deviation
```

```
[1] 11.31702
```

```
var(translations$price, na.rm = T)  # calculate the variance
```

```
[1] 128.075
```

We might also want to find the median and interquartile range.

```r
median(translations$price, na.rm = T)  #calculate the median
```

```
[1] 17
```

Three ways to find the interquartile range:

```r
# first way: calculate each quantile separately first then,
# take the difference
quantile(translations$price, na.rm = T, p = 0.75)  # .75 quantile
```

```
75%
 24
```

```r
quantile(translations$price, na.rm = T, p = 0.25)  # .25 quantile
```

```
25%
 15
```

```r
quantile(translations$price, na.rm = T, p = 0.75) - quantile(translations$price,
    na.rm = T, p = 0.25)
```

```
75%
  9
```

```r
# you can use this method to calculate the difference between
# any two quantiles, not just the 1st and the 3rd quartiles
```

```r
# alternatively, we can output the values of the two
# quantiles as elements in a vector and then take the
# difference of the two quantiles
diff(quantile(translations$price, na.rm = T, p = c(0.25, 0.75)))
```

```
75%
  9
```

```r
# lastly, we can also calculate this with the `IQR()`
# function
IQR(translations$price, na.rm = T)
```

```
[1] 9
```

Check for understanding

We can display the full five-number summary of `price` with one line of code.

```r
summary(translations$price)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
```

```
    0.0     15.0     17.0     20.4     24.0    282.0       14
```

With the outputs above answer the following questions.

Question 1. Roughly what percentage of the translations cost between 15 and 24 dollars?

_____

Answer: About 50%.

_____

Question 2. Roughly what percentage of translations cost more than 24 dollars?

_____

Answer: 100% - 75% = 25%.

_____

Question 3. Suppose that about 67.5% of the translations cost less than `x` dollars; `x` is called the _____ quantile. Find out what price this quantile corresponds to.

_____

Answer: The price that about 67.5% of the translations cost less than is called the .675 quantile. The price this .675 quantile corresponds to is 20 dollars.

```r
quantile(translations$price, p = 0.675, na.rm = T)
```

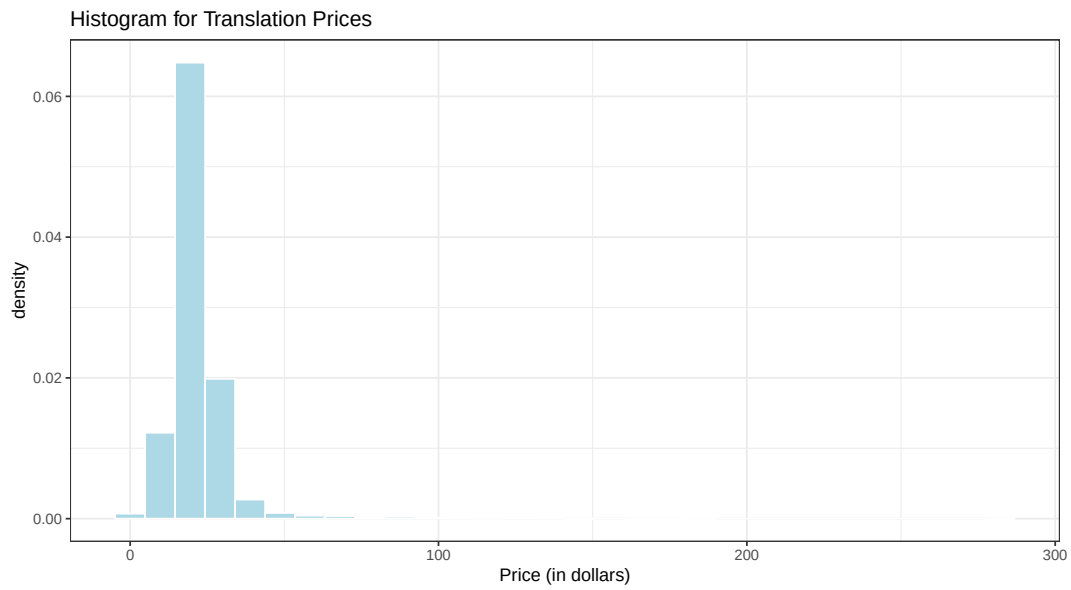
```
67.5%
   20
```

_____

Question 4. Let's look at the histogram for translation prices.

_____

Answer:



```
max(new_translations$price, na.rm = T)  # some translations seem to be a lot more expensive than
```

```
[1] 282
```

We see that some translations cost a lot more than others. Which translations are these? We can subset part of the data that pertains to the highest-cost translations. From the histogram, it looks like there are few books that cost more than 100 dollars. Let's see how many there are.

```
nrow(translations %>%
    filter(price > 100))
```

```
[1] 27
```

We can also sort the book prices in decreasing order:

```
translations %>%
    arrange(desc(price)) %>%
    pull(price)
```

```
  [1] 282.000 185.000 180.000 179.000 163.000 150.000 141.000 139.000
  [9] 135.000 135.000 134.950 130.000 130.000 129.000 125.000 125.000
 [17] 125.000 125.000 120.000 120.000 115.000 112.460 110.000 110.000
 [25] 110.000 109.950 104.950 100.000 100.000 100.000  99.000  95.000
 [33]  95.000  95.000  95.000  95.000  95.000  90.000  90.000  90.000
 [41]  90.000  90.000  90.000  90.000  90.000  89.950  89.950  89.950
```

[49]	89.950	89.000	88.950	85.000	85.000	85.000	85.000	85.000
[57]	85.000	85.000	85.000	85.000	85.000	80.000	80.000	79.950
[65]	79.950	79.950	78.000	76.000	75.000	75.000	75.000	75.000
[73]	75.000	75.000	74.950	74.950	74.950	74.500	74.000	70.000
[81]	70.000	70.000	70.000	69.950	69.950	69.950	69.950	69.950
[89]	69.000	68.950	68.000	65.000	65.000	65.000	65.000	65.000
[97]	65.000	65.000	65.000	65.000	65.000	65.000	65.000	65.000
[105]	65.000	65.000	65.000	65.000	64.950	60.000	60.000	60.000
[113]	60.000	60.000	60.000	59.950	59.950	59.950	59.950	59.950
[121]	59.950	59.950	59.950	59.950	59.000	59.000	55.000	55.000
[129]	55.000	55.000	55.000	55.000	55.000	55.000	55.000	55.000
[137]	55.000	55.000	55.000	55.000	55.000	55.000	54.950	54.950
[145]	54.000	52.500	50.000	50.000	50.000	50.000	50.000	50.000
[153]	50.000	50.000	50.000	50.000	50.000	50.000	50.000	50.000
[161]	50.000	50.000	50.000	50.000	50.000	50.000	50.000	49.990
[169]	49.950	49.950	49.950	49.950	49.950	49.950	49.950	49.950
[177]	49.950	49.500	49.000	48.000	45.000	45.000	45.000	45.000
[185]	45.000	45.000	45.000	45.000	45.000	45.000	45.000	45.000
[193]	45.000	45.000	45.000	45.000	45.000	45.000	45.000	45.000
[201]	45.000	45.000	45.000	45.000	45.000	45.000	45.000	45.000
[209]	45.000	45.000	44.750	44.500	44.000	43.000	42.000	42.000
[217]	41.000	41.000	40.000	40.000	40.000	40.000	40.000	40.000
[225]	40.000	40.000	40.000	40.000	40.000	40.000	40.000	40.000
[233]	40.000	40.000	40.000	40.000	40.000	40.000	40.000	40.000
[241]	40.000	40.000	40.000	40.000	40.000	40.000	40.000	40.000
[249]	40.000	40.000	39.990	39.950	39.950	39.950	39.950	39.950
[257]	39.950	39.950	39.950	39.950	39.950	39.950	39.950	39.950
[265]	39.950	39.950	39.950	39.950	39.950	39.950	39.950	39.950
[273]	39.950	39.950	39.950	39.950	39.950	39.950	39.950	39.950
[281]	39.950	39.500	39.500	39.500	39.500	39.500	39.000	38.950
[289]	38.000	38.000	38.000	38.000	38.000	38.000	38.000	38.000
[297]	38.000	37.950	37.500	36.950	36.950	36.950	36.000	36.000
[305]	36.000	36.000	36.000	35.950	35.000	35.000	35.000	35.000
[313]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[321]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[329]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[337]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[345]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[353]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[361]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[369]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[377]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[385]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[393]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	35.000
[401]	35.000	35.000	35.000	35.000	35.000	35.000	35.000	34.950
[409]	34.950	34.950	34.950	34.950	34.950	34.950	34.950	34.950
[417]	34.950	34.950	34.950	34.950	34.950	34.950	34.950	34.950
[425]	34.950	34.950	34.950	34.950	34.950	34.950	34.950	34.950

[433]	34.950	34.950	34.950	34.950	34.950	34.950	34.800	34.750
[441]	34.000	34.000	34.000	34.000	33.500	33.150	33.000	33.000
[449]	33.000	32.990	32.990	32.950	32.950	32.950	32.950	32.950
[457]	32.950	32.500	32.500	32.500	32.500	32.500	32.500	32.500
[465]	32.250	32.000	32.000	32.000	32.000	32.000	32.000	32.000
[473]	32.000	32.000	31.000	31.000	31.000	30.990	30.500	30.000
[481]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[489]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[497]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[505]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[513]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[521]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[529]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[537]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[545]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[553]	30.000	30.000	30.000	30.000	30.000	30.000	30.000	30.000
[561]	30.000	30.000	30.000	29.990	29.990	29.990	29.990	29.990
[569]	29.990	29.990	29.990	29.990	29.990	29.990	29.990	29.990
[577]	29.990	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[585]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[593]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[601]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[609]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[617]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[625]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[633]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[641]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[649]	29.950	29.950	29.950	29.950	29.950	29.950	29.950	29.950
[657]	29.950	29.950	29.950	29.950	29.950	29.950	29.890	29.600
[665]	29.500	29.500	29.500	29.500	29.500	29.260	29.200	29.000
[673]	29.000	29.000	29.000	29.000	29.000	29.000	29.000	28.990
[681]	28.990	28.990	28.990	28.990	28.990	28.990	28.990	28.990
[689]	28.990	28.990	28.990	28.950	28.950	28.950	28.950	28.950
[697]	28.950	28.950	28.950	28.950	28.950	28.950	28.950	28.950
[705]	28.950	28.950	28.950	28.950	28.950	28.950	28.950	28.700
[713]	28.500	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[721]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[729]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[737]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[745]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[753]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[761]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[769]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[777]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	28.000
[785]	28.000	28.000	28.000	28.000	28.000	28.000	28.000	27.990
[793]	27.990	27.990	27.990	27.990	27.990	27.990	27.990	27.990
[801]	27.990	27.990	27.990	27.990	27.990	27.990	27.990	27.990
[809]	27.990	27.990	27.990	27.990	27.990	27.990	27.990	27.950

[817]	27.950	27.950	27.950	27.950	27.950	27.950	27.950	27.950
[825]	27.950	27.950	27.950	27.950	27.950	27.950	27.950	27.950
[833]	27.950	27.950	27.950	27.950	27.950	27.950	27.950	27.950
[841]	27.950	27.950	27.950	27.950	27.950	27.950	27.950	27.950
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[1041]	26.990	26.990	26.990	26.990	26.990	26.990	26.990	26.990
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[2545]	22.000	22.000	22.000	22.000	22.000	22.000	22.000	22.000
[2553]	22.000	21.990	21.990	21.990	21.990	21.990	21.990	21.990
[2561]	21.990	21.990	21.990	21.990	21.990	21.990	21.990	21.990
[2569]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2577]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2585]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2593]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2601]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2609]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2617]	21.950	21.950	21.950	21.950	21.950	21.950	21.950	21.950
[2625]	21.950	21.950	21.500	21.500	21.500	21.500	21.500	21.500
[2633]	21.500	21.500	21.500	21.500	21.500	21.500	21.500	21.500
[2641]	21.500	21.500	21.500	21.500	21.500	21.500	21.500	21.500
[2649]	21.500	21.500	21.500	21.500	21.500	21.480	21.300	21.300
[2657]	21.240	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2665]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2673]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2681]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2689]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2697]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2705]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2713]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2721]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000
[2729]	21.000	21.000	21.000	21.000	21.000	21.000	21.000	21.000

[2737]	21.000	21.000	21.000	21.000	20.990	20.990	20.990	20.990
[2745]	20.990	20.950	20.950	20.950	20.950	20.950	20.750	20.210
[2753]	20.150	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2761]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2769]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2777]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2785]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2793]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2801]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2809]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2817]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2825]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2833]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2841]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2849]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2857]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2865]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2873]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2881]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2889]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2897]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2905]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2913]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2921]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2929]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2937]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2945]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2953]	20.000	20.000	20.000	20.000	20.000	20.000	20.000	20.000
[2961]	20.000	20.000	20.000	20.000	20.000	20.000	19.990	19.990
[2969]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[2977]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[2985]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[2993]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[3001]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[3009]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[3017]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[3025]	19.990	19.990	19.990	19.990	19.990	19.990	19.990	19.990
[3033]	19.990	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3041]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3049]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3057]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3065]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3073]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3081]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3089]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3097]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3105]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3113]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950

[3121]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3129]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3137]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3145]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3153]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3161]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3169]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3177]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3185]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3193]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3201]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3209]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3217]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3225]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3233]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3241]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3249]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3257]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3265]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3273]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3281]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3289]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3297]	19.950	19.950	19.950	19.950	19.950	19.950	19.950	19.950
[3305]	19.950	19.950	19.950	19.950	19.900	19.800	19.800	19.500
[3313]	19.500	19.500	19.500	19.500	19.500	19.500	19.500	19.260
[3321]	19.200	19.200	19.000	19.000	19.000	19.000	19.000	19.000
[3329]	19.000	19.000	19.000	19.000	19.000	19.000	19.000	19.000
[3337]	19.000	19.000	19.000	19.000	19.000	19.000	19.000	19.000
[3345]	19.000	19.000	19.000	19.000	19.000	19.000	19.000	19.000
[3353]	19.000	19.000	19.000	19.000	19.000	19.000	19.000	19.000
[3361]	19.000	19.000	19.000	19.000	19.000	19.000	18.990	18.990
[3369]	18.990	18.990	18.990	18.990	18.990	18.990	18.990	18.990
[3377]	18.990	18.990	18.990	18.990	18.990	18.990	18.990	18.990
[3385]	18.990	18.990	18.990	18.990	18.990	18.990	18.990	18.990
[3393]	18.990	18.990	18.990	18.990	18.990	18.990	18.950	18.950
[3401]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3409]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3417]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3425]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3433]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3441]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3449]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3457]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3465]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3473]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3481]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3489]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3497]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950

[3505]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3513]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3521]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3529]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3537]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3545]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3553]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3561]	18.950	18.950	18.950	18.950	18.950	18.950	18.950	18.950
[3569]	18.950	18.950	18.950	18.950	18.950	18.900	18.860	18.630
[3577]	18.500	18.500	18.500	18.500	18.500	18.500	18.500	18.500
[3585]	18.500	18.500	18.095	18.000	18.000	18.000	18.000	18.000
[3593]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3601]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3609]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3617]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3625]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3633]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3641]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3649]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3657]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3665]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3673]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3681]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3689]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3697]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3705]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3713]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3721]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3729]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3737]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3745]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3753]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3761]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3769]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3777]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3785]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3793]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3801]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3809]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3817]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3825]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3833]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3841]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3849]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3857]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3865]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3873]	18.000	18.000	18.000	18.000	18.000	18.000	18.000	18.000
[3881]	18.000	17.990	17.990	17.990	17.990	17.990	17.990	17.990

[3889]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3897]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3905]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3913]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3921]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3929]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3937]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3945]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3953]	17.990	17.990	17.990	17.990	17.990	17.990	17.990	17.990
[3961]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[3969]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[3977]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[3985]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[3993]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4001]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4009]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4017]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4025]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4033]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4041]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4049]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4057]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4065]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4073]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4081]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4089]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4097]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4105]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4113]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4121]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4129]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4137]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4145]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4153]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4161]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4169]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4177]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4185]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4193]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4201]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4209]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.950
[4217]	17.950	17.950	17.950	17.950	17.950	17.950	17.950	17.840
[4225]	17.750	17.500	17.500	17.500	17.500	17.500	17.500	17.000
[4233]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4241]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4249]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4257]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4265]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000

[4273]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4281]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4289]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4297]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4305]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4313]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4321]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4329]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4337]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4345]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4353]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4361]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4369]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4377]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4385]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4393]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4401]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4409]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4417]	17.000	17.000	17.000	17.000	17.000	17.000	17.000	17.000
[4425]	17.000	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4433]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4441]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4449]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4457]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4465]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4473]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4481]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4489]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4497]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4505]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4513]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4521]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.990
[4529]	16.990	16.990	16.990	16.990	16.990	16.990	16.990	16.960
[4537]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4545]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4553]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4561]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4569]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4577]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4585]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4593]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4601]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4609]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4617]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4625]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4633]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4641]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4649]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950

[4657]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4665]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4673]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4681]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4689]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4697]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4705]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4713]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4721]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4729]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4737]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4745]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4753]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4761]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4769]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4777]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4785]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4793]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4801]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4809]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4817]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4825]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4833]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4841]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4849]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4857]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4865]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4873]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4881]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4889]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4897]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4905]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4913]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4921]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4929]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4937]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4945]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4953]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4961]	16.950	16.950	16.950	16.950	16.950	16.950	16.950	16.950
[4969]	16.950	16.950	16.750	16.500	16.500	16.500	16.500	16.500
[4977]	16.500	16.500	16.500	16.500	16.500	16.120	16.000	16.000
[4985]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[4993]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
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[5009]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5017]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5025]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5033]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000

[5041]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5049]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5057]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5065]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5073]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5081]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5089]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5097]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
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[5113]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5121]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5129]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5137]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5145]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5153]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5161]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5169]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5177]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5185]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5193]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5201]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5209]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5217]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5225]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5233]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5241]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5249]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5257]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5265]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5273]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5281]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5289]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5297]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5305]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5313]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5321]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5329]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5337]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5345]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5353]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5361]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5369]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
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[5385]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5393]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
[5401]	16.000	16.000	16.000	16.000	16.000	16.000	16.000	16.000
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[5417]	16.000	16.000	16.000	16.000	16.000	16.000	15.990	15.990

[5425]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5433]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5441]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5449]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5457]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5465]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
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[5513]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5521]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5529]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5537]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5545]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.990
[5553]	15.990	15.990	15.990	15.990	15.990	15.990	15.990	15.980
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[5577]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
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[5601]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5609]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5617]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5625]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5633]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5641]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5649]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5657]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5665]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5673]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5681]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5689]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5697]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5705]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5713]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5721]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5729]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5737]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5745]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5753]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5761]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5769]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5777]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5785]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5793]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5801]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950

[5809]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5817]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5825]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5833]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5841]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5849]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5857]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5865]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5873]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5881]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5889]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5897]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5905]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5913]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5921]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5929]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5937]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5945]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5953]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5961]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5969]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5977]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5985]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[5993]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6001]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6009]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6017]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6025]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6033]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6041]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6049]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6057]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6065]	15.950	15.950	15.950	15.950	15.950	15.950	15.950	15.950
[6073]	15.950	15.950	15.950	15.500	15.500	15.500	15.500	15.500
[6081]	15.500	15.500	15.500	15.500	15.500	15.500	15.490	15.000
[6089]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6097]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6105]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6113]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6121]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6129]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6137]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6145]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6153]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6161]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6169]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6177]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6185]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000

[6193]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6201]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6209]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6217]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6225]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6233]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6241]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6249]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6257]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6265]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6273]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6281]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6289]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6297]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6305]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6313]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6321]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6329]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6337]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6345]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6353]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6361]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6369]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6377]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6385]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6393]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6401]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6409]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6417]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6425]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6433]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6441]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6449]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6457]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6465]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6473]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6481]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6489]	15.000	15.000	15.000	15.000	15.000	15.000	15.000	15.000
[6497]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6505]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6513]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6521]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6529]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6537]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6545]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6553]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6561]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6569]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990

[6577]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.990
[6585]	14.990	14.990	14.990	14.990	14.990	14.990	14.990	14.950
[6593]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6601]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6609]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6617]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6625]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6633]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6641]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6649]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6657]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6665]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6673]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6681]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6689]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6697]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6705]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6713]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6721]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6729]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6737]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6745]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6753]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6761]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6769]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6777]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6785]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6793]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6801]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6809]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6817]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6825]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6833]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6841]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6849]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6857]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6865]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6873]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6881]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6889]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6897]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6905]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6913]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6921]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6929]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6937]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6945]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6953]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950

[6961]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6969]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6977]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6985]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[6993]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7001]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7009]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7017]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7025]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7033]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7041]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7049]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7057]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7065]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7073]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7081]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7089]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7097]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7105]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7113]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7121]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7129]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7137]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7145]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7153]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7161]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7169]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7177]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7185]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7193]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7201]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7209]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7217]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7225]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7233]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7241]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7249]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7257]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7265]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7273]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7281]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7289]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7297]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7305]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7313]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7321]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7329]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7337]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950

[7345]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7353]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7361]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7369]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7377]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7385]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7393]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7401]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7409]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7417]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7425]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7433]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7441]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7449]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7457]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7465]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7473]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7481]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7489]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7497]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7505]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7513]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7521]	14.950	14.950	14.950	14.950	14.950	14.950	14.950	14.950
[7529]	14.950	14.950	14.950	14.500	14.500	14.500	14.500	14.500
[7537]	14.500	14.500	14.100	14.000	14.000	14.000	14.000	14.000
[7545]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7553]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7561]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7569]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7577]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7585]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7593]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7601]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7609]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7617]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7625]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7633]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7641]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7649]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7657]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7665]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7673]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7681]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7689]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7697]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7705]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	14.000
[7713]	14.000	14.000	14.000	14.000	14.000	14.000	14.000	13.990
[7721]	13.990	13.990	13.990	13.990	13.990	13.990	13.990	13.990

[7729]	13.990	13.990	13.990	13.990	13.990	13.990	13.990	13.990
[7737]	13.990	13.990	13.990	13.990	13.990	13.990	13.990	13.990
[7745]	13.990	13.990	13.990	13.990	13.990	13.990	13.990	13.990
[7753]	13.990	13.990	13.990	13.950	13.950	13.950	13.950	13.950
[7761]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7769]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7777]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7785]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7793]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7801]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7809]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7817]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7825]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7833]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7841]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7849]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7857]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7865]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7873]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7881]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7889]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7897]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7905]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7913]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7921]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7929]	13.950	13.950	13.950	13.950	13.950	13.950	13.950	13.950
[7937]	13.950	13.950	13.950	13.850	13.500	13.500	13.500	13.500
[7945]	13.500	13.500	13.500	13.500	13.500	13.500	13.500	13.500
[7953]	13.500	13.450	13.000	13.000	13.000	13.000	13.000	13.000
[7961]	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000
[7969]	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000
[7977]	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000
[7985]	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000
[7993]	13.000	13.000	13.000	13.000	13.000	13.000	13.000	13.000
[8001]	13.000	12.990	12.990	12.990	12.990	12.990	12.990	12.990
[8009]	12.990	12.990	12.990	12.990	12.990	12.990	12.990	12.990
[8017]	12.990	12.990	12.990	12.990	12.990	12.990	12.990	12.990
[8025]	12.990	12.990	12.990	12.990	12.990	12.990	12.990	12.990
[8033]	12.990	12.990	12.990	12.990	12.990	12.980	12.950	12.950
[8041]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8049]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8057]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8065]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8073]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8081]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8089]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8097]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8105]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950

[8113]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8121]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8129]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8137]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8145]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8153]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8161]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8169]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8177]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8185]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8193]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8201]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8209]	12.950	12.950	12.950	12.950	12.950	12.950	12.950	12.950
[8217]	12.950	12.800	12.500	12.500	12.500	12.500	12.500	12.500
[8225]	12.500	12.500	12.500	12.500	12.500	12.500	12.500	12.000
[8233]	12.000	12.000	12.000	12.000	12.000	12.000	12.000	12.000
[8241]	12.000	12.000	12.000	12.000	12.000	12.000	12.000	12.000
[8249]	12.000	12.000	12.000	12.000	12.000	12.000	12.000	12.000
[8257]	12.000	12.000	12.000	12.000	12.000	12.000	12.000	12.000
[8265]	12.000	12.000	12.000	12.000	12.000	12.000	12.000	12.000
[8273]	12.000	12.000	12.000	12.000	12.000	12.000	12.000	12.000
[8281]	12.000	12.000	12.000	12.000	12.000	11.990	11.990	11.990
[8289]	11.990	11.990	11.990	11.990	11.990	11.990	11.990	11.990
[8297]	11.990	11.950	11.950	11.950	11.950	11.950	11.950	11.950
[8305]	11.950	11.950	11.950	11.950	11.950	11.950	11.950	11.950
[8313]	11.950	11.950	11.950	11.950	11.950	11.950	11.400	11.370
[8321]	11.150	11.060	11.060	11.000	11.000	11.000	11.000	11.000
[8329]	11.000	11.000	10.990	10.990	10.990	10.990	10.990	10.990
[8337]	10.990	10.990	10.990	10.990	10.990	10.990	10.990	10.990
[8345]	10.990	10.990	10.990	10.990	10.990	10.990	10.990	10.990
[8353]	10.990	10.990	10.990	10.990	10.990	10.990	10.990	10.990
[8361]	10.990	10.990	10.990	10.950	10.950	10.950	10.950	10.950
[8369]	10.950	10.950	10.950	10.950	10.950	10.950	10.950	10.950
[8377]	10.950	10.950	10.950	10.950	10.950	10.950	10.950	10.950
[8385]	10.950	10.950	10.950	10.950	10.950	10.690	10.500	10.000
[8393]	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
[8401]	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
[8409]	10.000	10.000	10.000	10.000	9.995	9.990	9.990	9.990
[8417]	9.990	9.990	9.990	9.990	9.990	9.990	9.990	9.990
[8425]	9.990	9.990	9.990	9.990	9.990	9.990	9.990	9.990
[8433]	9.990	9.990	9.990	9.990	9.990	9.950	9.950	9.950
[8441]	9.950	9.950	9.950	9.950	9.950	9.950	9.950	9.950
[8449]	9.950	9.950	9.950	9.950	9.950	9.950	9.950	9.950
[8457]	9.950	9.950	9.950	9.950	9.950	9.950	9.950	9.950
[8465]	9.950	9.660	9.500	9.370	9.250	9.000	9.000	9.000
[8473]	9.000	9.000	9.000	8.990	8.990	8.990	8.990	8.990
[8481]	8.990	8.990	8.950	8.950	8.950	8.950	8.700	8.180
[8489]	8.000	8.000	7.990	7.990	7.990	7.990	7.990	7.990

```
[8497]   7.990   7.990   7.990   7.990   7.990   7.990   7.990   7.950
[8505]   7.950   7.950   7.950   7.950   7.950   7.950   7.890   7.760
[8513]   7.280   7.280   7.280   7.280   6.990   6.990   6.990   6.990
[8521]   6.950   6.950   6.950   6.950   6.950   6.950   6.950   6.950
[8529]   6.950   6.310   6.310   6.310   6.210   5.990   5.990   5.990
[8537]   5.990   5.990   5.990   5.990   5.990   5.990   5.990   5.950
[8545]   5.950   5.820   5.820   5.500   5.100   5.000   4.990   4.990
[8553]   4.990   4.950   4.360   3.990   3.990   2.990   2.990   2.990
[8561]   2.990   2.990   1.990   1.990   1.990   1.990   1.990   1.990
[8569]   1.990   1.990   1.990   1.990   1.990   1.990   1.990   1.990
[8577]   1.990   1.990   1.990   1.990   1.990   1.990   1.990   1.990
[8585]   1.990   1.990   1.990   0.990   0.990   0.990   0.990   0.990
[8593]   0.990   0.990   0.000   0.000   0.000   0.000   0.000   0.000
[8601]   0.000   0.000   0.000   0.000   0.000   0.000   0.000   0.000
[8609]   0.000   0.000   0.000   0.000   0.000   0.000      NA      NA
[8617]      NA      NA      NA      NA      NA      NA      NA      NA
[8625]      NA      NA      NA      NA
```

The list above is too long; in general you don't want to print out figures that are irrelevant to your analysis. Let's look at just the top 20 prices.

```r
# print only the first 20 lines
translations %>%
    arrange(desc(price)) %>%
    pull(price) %>%
    head(n = 20)
```

```
 [1] 282.00 185.00 180.00 179.00 163.00 150.00 141.00 139.00 135.00
[10] 135.00 134.95 130.00 130.00 129.00 125.00 125.00 125.00 125.00
[19] 120.00 120.00
```

Another way to do this is:

```r
translations %>%
    slice_max(order_by = price, n = 20) %>%
    pull(price)
```

```
 [1] 282.00 185.00 180.00 179.00 163.00 150.00 141.00 139.00 135.00
[10] 135.00 134.95 130.00 130.00 129.00 125.00 125.00 125.00 125.00
[19] 120.00 120.00
```

Note that the most expensive translation is almost 100 dollars more than the second highest one! Why might these books be so expensive? Perhaps they are rare, limited-edition prints. Or perhaps translations from some languages are more expensive than others. It's also possible that these could be input errors. We should study the dataset more to come to a proper conclusion.

A price that falls well outside the main price range is called an outlier; we are almost certain that these top prices are outliers just from looking at the histogram.

Outliers

An **Outlier** is an observation that falls well outside the main range of the data. We should treat outliers with caution; sometimes outliers occur due to recording errors and other times outliers actually provide us insightful information about the data.

There are different rules to identify outliers. One set of rules that is commonly used defines an outlier to be a data point that falls more than 1.5 times the IQR below Q1 or above the Q3.

```
quantile(translations$price, na.rm = T)  #show all quartiles
```

```
  0%  25%  50%  75% 100%
   0   15   17   24  282
```

```
quartiles <- quantile(translations$price, na.rm = T)

# Use index to reference the proper quartile
quartiles[2] - 1.5 * IQR(translations$price, na.rm = T)  # lower bound
```

```
25%
1.5
```

```
quartiles[4] + 1.5 * IQR(translations$price, na.rm = T)  # upper bound
```

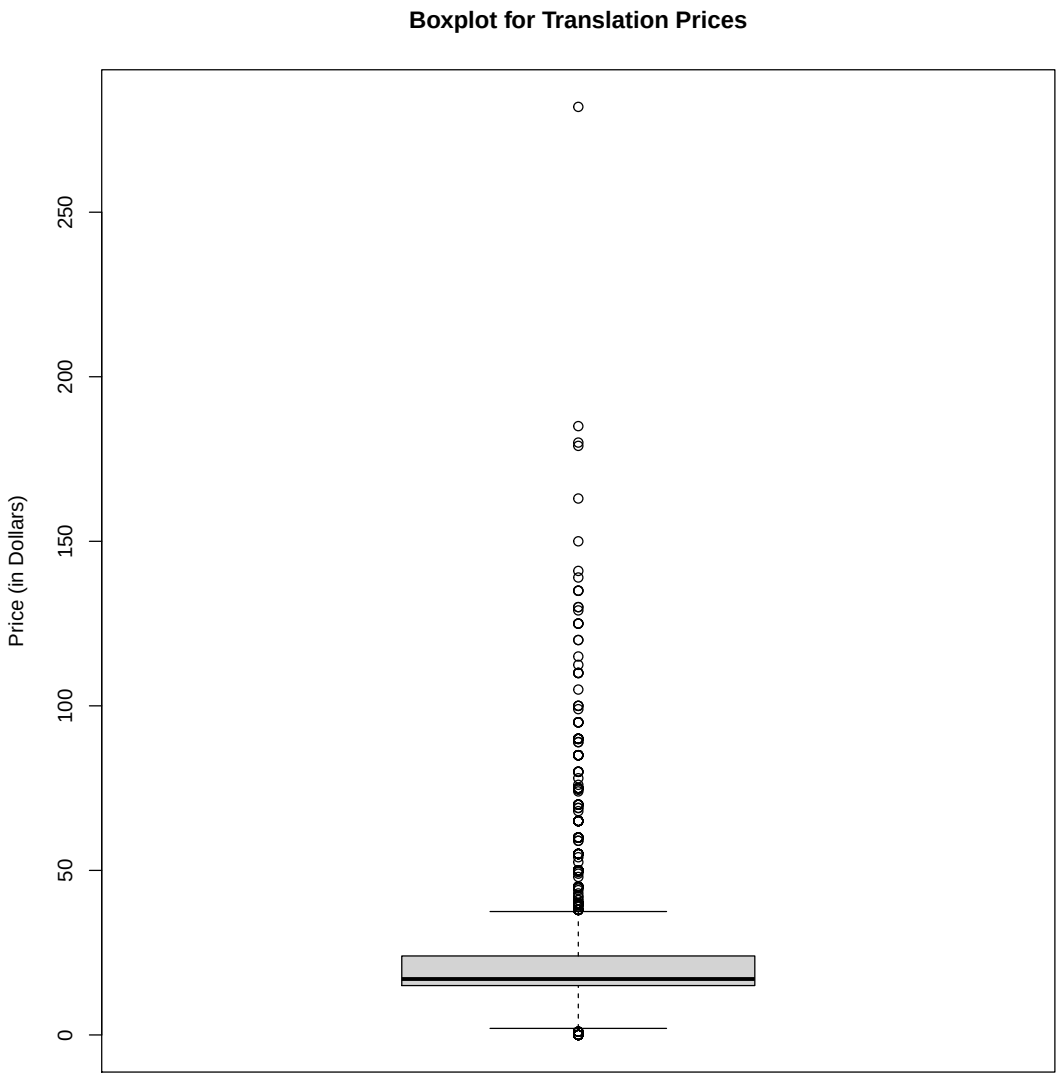
```
 75%
37.5
```

```
lower <- quartiles[2] - 1.5 * IQR(translations$price, na.rm = T)  # lower bound
upper <- quartiles[4] + 1.5 * IQR(translations$price, na.rm = T)  # upper bound

translations %>%
    filter(price < lower | price > upper) %>%
    nrow()  #calculate number of outliers
```

```
[1] 325
```

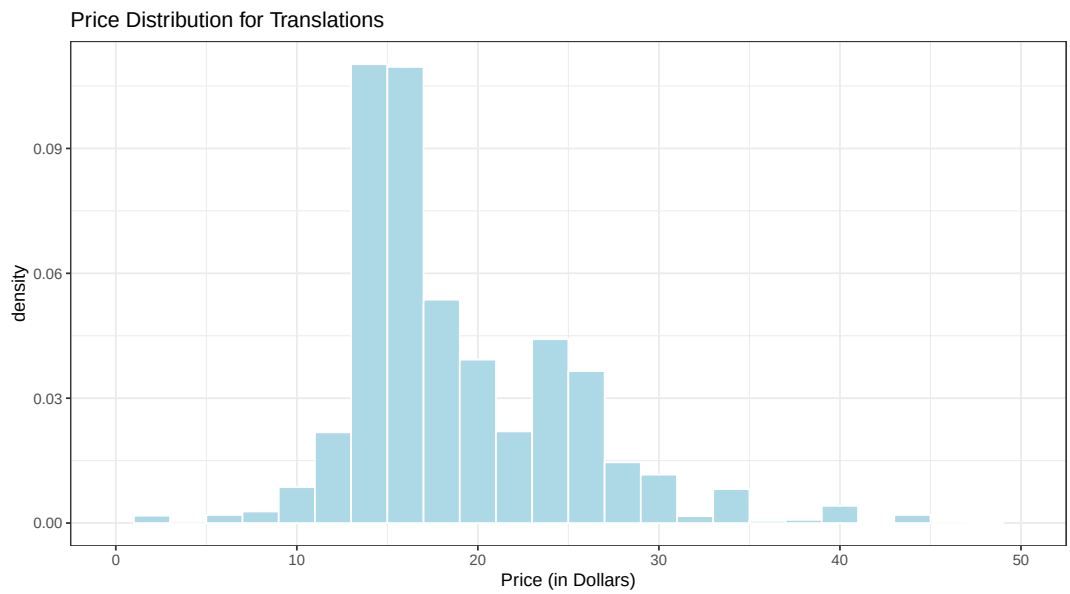
In addition, a boxplot provides a graphical way to identify outliers.



Mode

The **modes** of a dataset are the peaks on the histogram for the dataset. The location of the mode tells us on which part of the x-interval the data has high frequency. The most prominent mode is called the **major mode**–there could be more than one major mode depending on the distribution. The less prominent ones are called the minor modes.

Question 5. Observe the following modified histogram for `price`, which shows only translations costing between 0 and 50 dollars; are there any modes?



Answer: There is one major mode around 14 dollars and one minor mode around 24 dollars.

Question 6. Exactly how many translations have a price within the interval (13,15]? (Hint: When you apply mathematical operations on a vector with logical arguments `R` treats logical arguments as numbers; `TRUE` is treated as 1 and `False` is treated as 0).

```
sum(translations$price > 13 & translations$price <= 15, na.rm = T)
```

```
[1] 1867
```

What is the proportion of translations with a price in the interval (13,15]?

```
mean(translations$price > 13 & translations$price <= 15, na.rm = T)
```

```
[1] 0.2167402
```

`geom_histogram()` uses `stat_bin()` to divide the data into bins. The default value for stat_bin() is `closed = "right"`. This means that the bin covers the value on its right-end-point but not the left-end-point.

Use the histogram to approximate the proportion of passengers are in the age interval (13, 15].

```r
0.1125 * 2  #observe the height and the width of the corresponding bar
```

```
[1] 0.225
```

- How much is the most expensive translation?
- How much is the least expensive translation?

```r
range(translations$price, na.rm = T)
```

```
[1]   0 282
```

Question 7. What are the prices of the four most expensive translations?

```r
# sort price in decreasing order
translations %>%
  slice_min(order_by = -price, n = 20) %>% #note that -price tells R to sort in descending order
  pull(price)
```

```
 [1] 282.00 185.00 180.00 179.00 163.00 150.00 141.00 139.00 135.00
[10] 135.00 134.95 130.00 130.00 129.00 125.00 125.00 125.00 125.00
[19] 120.00 120.00
```

The four most expensive translations in the database cost:

```r
translations %>%
    slice_min(order_by = -price, n = 4) %>%
    pull(price)
```

```
[1] 282 185 180 179
```

Additional Practice

Question 1. Create the following objects in R.

(1) A data frame with translation title, author's first and last name, translator's first and last name, publication year, author's gender and translator's gender.

```r
translations %>%
    select(title, auth_first, auth_last, trnsl_first, trnsl_last,
        pubdate_yr, auth_gend, trnsl_gend)
```

```
# A tibble: 8,628 x 8
   title        auth_first auth_last trnsl_first trnsl_last pubdate_yr
   <chr>        <chr>      <chr>     <chr>       <chr>           <dbl>
 1 Equatoria    Tom        Dreyer    Michiel     Heyns            2008
 2 Blood of th~ Azem       Shkreli   Robert      Elsie            2008
 3 Dreaming Es~ Valentina  Saracini  Erica       Weitzman         2008
 4 Taxi         Khaled Al  Khamissi  Jonathan    Wright           2008
 5 Final Bet    Abdelilah  Hamdouchi Jonathan    Smolin           2008
 6 As Doha Said Bahaa      Taher     Peter       Daniel           2008
 7 Final Night  Buthaina ~ Nasiri    Denys       Johnson-D~       2008
 8 Cell Block ~ Fadhil Al  Azzawi    William M.  Hutchins         2008
 9 Crane        Halim      Barakat   Bassam      Frangieh         2008
10 Gold Dust    Ibrahim Al Koni      Elliott     Colla            2008
# ... with 8,618 more rows, and 2 more variables: auth_gend <chr>,
#   trnsl_gend <chr>
```

Or alternatively:

```
translations %>%
    select(title, starts_with("auth"), starts_with("trnsl"),
        pubdate_yr)
```

```
# A tibble: 8,628 x 8
   title         auth_first auth_last auth_gend trnsl_first trnsl_last
   <chr>         <chr>      <chr>     <chr>     <chr>       <chr>
 1 Equatoria     Tom        Dreyer    Male      Michiel     Heyns
 2 Blood of the~ Azem       Shkreli   Male      Robert      Elsie
 3 Dreaming Esc~ Valentina  Saracini  Female    Erica       Weitzman
 4 Taxi          Khaled Al  Khamissi  Male      Jonathan    Wright
 5 Final Bet     Abdelilah  Hamdouchi Male      Jonathan    Smolin
 6 As Doha Said  Bahaa      Taher     Male      Peter       Daniel
 7 Final Night   Buthaina ~ Nasiri    Female    Denys       Johnson-D~
 8 Cell Block F~ Fadhil Al  Azzawi    Male      William M.  Hutchins
 9 Crane         Halim      Barakat   Male      Bassam      Frangieh
10 Gold Dust     Ibrahim Al Koni      Male      Elliott     Colla
# ... with 8,618 more rows, and 2 more variables: trnsl_gend <chr>,
#   pubdate_yr <dbl>
```

---

(2) A dataframe with only translations from Germany or France.

```
translations %>%
    filter(country == "Germany" | country == "France") %>%
    arrange(country)  #list by country in alphabetical order
```

```
# A tibble: 1,995 x 18
       isbn `e-isbn` title auth_first auth_last trnsl_first trnsl_last
      <dbl> <lgl>    <chr> <chr>      <chr>     <chr>       <chr>
 1  9.78e12 NA       Pari~ Aurelien   Masson    various     various
 2  9.78e12 NA       Wait~ Dominique  Fabre     Jordan      Stump
```

```
 3  9.78e12 NA       Smal~ Pierre     Michon    Jody        Gladding
 4  9.78e12 NA       Art ~ Eugene     Guillevic Maureen     Smith
 5  9.78e12 NA       Esse~ Joyce      Mansour   Serge       Gavronsky
 6  9.78e12 NA       EyeS~ Raymond    Queneau   Daniela     Hurezanu
 7  9.78e12 NA       Esse~ Robert     Desnos    Mary Ann    Caws
 8  9.78e12 NA       Poem~ Valery     Larbaud   Ron         Padgett
 9  9.78e12 NA       You ~ Caroline   Dubois    Cole        Swensen
10  9.78e12 NA       Secr~ Isabelle ~ Howald    Elena       Rivera
# ... with 1,985 more rows, and 11 more variables: publisher <chr>,
#   genre <chr>, price <dbl>, pubdate_mo <chr>, pubdate_yr <dbl>,
#   language <chr>, country <chr>, addl_auth <chr>, add_trnsl <chr>,
#   auth_gend <chr>, trnsl_gend <chr>
```

---

(3) A vector that includes all of the unique languages in the translation database.

```
translations %>%
    pull(language) %>%
    unique()
```

```
 [1] "Afrikaans"   "Albanian"    "Arabic"      "Basque"
 [5] "Belarussian" "Bengali"     "Bosnian"     "Bulgarian"
 [9] "Burmese"     "Catalan"     "Chinese"     "Croatian"
[13] "Czech"       "Danish"      "Dutch"       "Estonian"
[17] "Finnish"     "French"      "Galician"    "German"
[21] "Greek"       "Hebrew"      "Hindi"       "Hungarian"
[25] "Icelandic"   "Italian"     "Japanese"    "Korean"
[29] "Norwegian"   "Persian"     "Polish"      "Portuguese"
[33] "Romanian"    "Russian"     "Serbian"     "Slovenian"
[37] "Spanish"     "Swedish"     "Tamil"       "Turkish"
[41] "Ukrainian"   "Urdu"        "Various"     "Vietnamese"
[45] "Welsh"       "Yiddish"     "Zulu"        "Ladino"
[49] "Latin"       "Lithuanian"  "Dari"        "Flemish"
[53] "Macedonian"  "Faroese"     "Gaelic"      "Georgian"
[57] "Pashto"      "Tigrinya"    NA            "Armenian"
[61] "Azerbaijani" "Indonesian"  "Innu-aimun"  "Irish"
[65] "Latvian"     "Marathi"     "Mayan"       "Slovak"
[69] "Zapotec"     "Anishinaabe" "Gujarati"    "Swahili"
[73] "Uyghur"      "Kurdish"     "Malayali"    "Mongolian"
[77] "Wolof"       "Esperanto"   "Guarani"     "Kannada"
[81] "Lingala"     "Occitan"     "Thai"        "Farsi"
```

How many unique languages are there?

```
length(translations %>%
    pull(language) %>%
    unique())
```

```
[1] 84
```

If we're being super attentive, we'll actually notice that one of the languages is listed as NA (i.e., the language was not listed for that observation). So there are actually 83 unique languages in the dataset- details matter!

Question 2. Create a table that shows the number of female and male translators by country. For now you can ignore observations for which the translator's gender is not provided.

```
translations %>%
    group_by(country) %>%
    summarize(ct.female = sum(trnsl_gend == "Female", na.rm = T),
        ct.male = sum(trnsl_gend == "Male", na.rm = T))
```

```
# A tibble: 144 x 3
   country         ct.female ct.male
   <chr>               <int>   <int>
 1 ---------------         0       0
 2 Afghanistan             4       2
 3 Albania                 5      10
 4 Algeria                12      21
 5 Angola                  0       7
 6 Argentina             139      77
 7 Armenia                 4       5
 8 Australia               0       1
 9 Austria                45      58
10 Azerbaijan              3       0
# ... with 134 more rows
```

Question 3. Modify one function from your answer to the previous question to instead show the *proportion* of female and male translators by country.

```
# Table with proportion of female and male translators by
# country
translations %>%
    group_by(country) %>%
    summarize(prp.female = mean(trnsl_gend == "Female", na.rm = T),
        prp.male = mean(trnsl_gend == "Male", na.rm = T))
```

```
# A tibble: 144 x 3
   country         prp.female prp.male
   <chr>                <dbl>    <dbl>
 1 ---------------      0        0
 2 Afghanistan          0.571    0.286
 3 Albania              0.278    0.556
 4 Algeria              0.353    0.618
 5 Angola               0        1
 6 Argentina            0.602    0.333
 7 Armenia              0.364    0.455
```

```
 8 Australia            0        0.5
 9 Austria              0.421    0.542
10 Azerbaijan           1        0
# ... with 134 more rows
```

We could easily modify our code to show the proportions as percentages:

```r
# Table with proportion of female and male translators by
# country
translations %>%
    group_by(country) %>%
    summarize(pct.female = mean(trnsl_gend == "Female", na.rm = T) *
        100, pct.male = mean(trnsl_gend == "Male", na.rm = T) *
        100)
```

```
# A tibble: 144 x 3
   country         pct.female pct.male
   <chr>                <dbl>    <dbl>
 1 ---------------        0        0
 2 Afghanistan           57.1     28.6
 3 Albania               27.8     55.6
 4 Algeria               35.3     61.8
 5 Angola                 0      100
 6 Argentina             60.2     33.3
 7 Armenia               36.4     45.5
 8 Australia              0       50
 9 Austria               42.1     54.2
10 Azerbaijan           100        0
# ... with 134 more rows
```

Question 4. Create a table to show which languages are most represented in the translation database.

```r
translations %>%
    group_by(language) %>%
    summarize(count = n(), percentage = round((n()/nrow(translations)) *
        100, digits = 1)) %>%
    arrange(desc(count)) %>%
    mutate(rank = rank(-count))  #Add column to show rank
```

```
# A tibble: 84 x 4
   language  count percentage  rank
   <chr>     <int>      <dbl> <dbl>
 1 French     1655       19.2     1
 2 Spanish    1171       13.6     2
 3 German     1051       12.2     3
 4 Italian     532        6.2     4
 5 Japanese    454        5.3     5
 6 Swedish     344        4       6
```

```
 7 Arabic      316        3.7     7
 8 Chinese     303        3.5     8
 9 Russian     287        3.3     9
10 Norwegian   223        2.6    10
# ... with 74 more rows
```

Change one part of your code to do the same for country, and then for publisher.

```
translations %>%
    group_by(country) %>%
    summarize(count = n(), percentage = round((n()/nrow(translations)) *
        100, digits = 1)) %>%
    arrange(desc(count)) %>%
    mutate(rank = rank(-count))  #Add column to show rank
```

```
# A tibble: 144 x 4
   country   count percentage  rank
   <chr>     <int>      <dbl> <dbl>
 1 France     1186       13.7     1
 2 Germany     809        9.4     2
 3 Italy       520        6       3
 4 Spain       464        5.4     4
 5 Japan       454        5.3     5
 6 Sweden      333        3.9     6
 7 China       281        3.3     7
 8 Russia      272        3.2     8
 9 Argentina   247        2.9     9
10 Norway      222        2.6    10
# ... with 134 more rows
```

```
translations %>%
    group_by(publisher) %>%
    summarize(count = n(), percentage = round((n()/nrow(translations)) *
        100, digits = 1)) %>%
    arrange(desc(count)) %>%
    mutate(rank = rank(-count))  #Add column to show rank
```

```
# A tibble: 884 x 4
   publisher       count percentage  rank
   <chr>           <int>      <dbl> <dbl>
 1 AmazonCrossing    439        5.1     1
 2 Dalkey Archive    324        3.8     2
 3 Seagull Books     226        2.6     3
 4 Europa Editions   200        2.3     4
 5 New Directions    182        2.1     5
 6 Other Press       145        1.7     6
 7 FSG               139        1.6     7
 8 Archipelago       120        1.4     8
 9 Penguin           119        1.4     9
10 Open Letter       115        1.3    10
```

```
# ... with 874 more rows
```

Question 5. Determine if each of the following countries is included in the translation database.

Brazil
Cyprus
Laos
Nigeria
Zimbabwe

You could print out a list of all countries included in the translation database and then verify individually, but there's a much simpler way. We can use the `%in%` function to verify this instantaneously.

```
"Brazil" %in% translations$country
```

```
[1] TRUE
```

```
"Cyprus" %in% translations$country
```

```
[1] TRUE
```

```
"Laos" %in% translations$country
```

```
[1] FALSE
```

```
"Nigeria" %in% translations$country
```

```
[1] FALSE
```

```
"Zimbabwe" %in% translations$country
```

```
[1] FALSE
```

We can write this more efficiently with a vector:

```
c("Brazil", "Cyprus", "Laos", "Nigeria", "Zimbabwe") %in% translations$country
```

```
[1]  TRUE  TRUE FALSE FALSE FALSE
```

For Discussion

Just as data is more than simply letters and numbers, data analysis is more than a computer at work. It relies on human interaction and interpretation. Consider the guiding questions below, to be discussed with your fellow classmates.

Guiding Question 1

Review the variables in the translation database. What information is missing that might be relevant to your analysis? What would you like to see in the dataset that isn't there?

Outline the processes (and potential roadblocks) to acquiring such information.

Guiding Question 2

Consider variables that may not fully represent all possible contributors. For example, what are the limitations to categorizing gender as binary (male or female)? How might the dataset better account for transgender writers and translators?

Guiding Question 3

Make a list of 3-5 questions that a researcher could try tackling using this dataset. Think big!

For example, how has the ratio of female to male translators changed over the last decade?